



Complex Interplay of Tandem, Segmental, Whole Genome Duplication, and Re-organization Drives Expansion of SAUR Gene Family in Brassicaceae

Richa Shukla¹ · Ekta Pokhriyal¹ · Sandip Das¹

Received: 3 November 2024 / Accepted: 10 June 2025

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

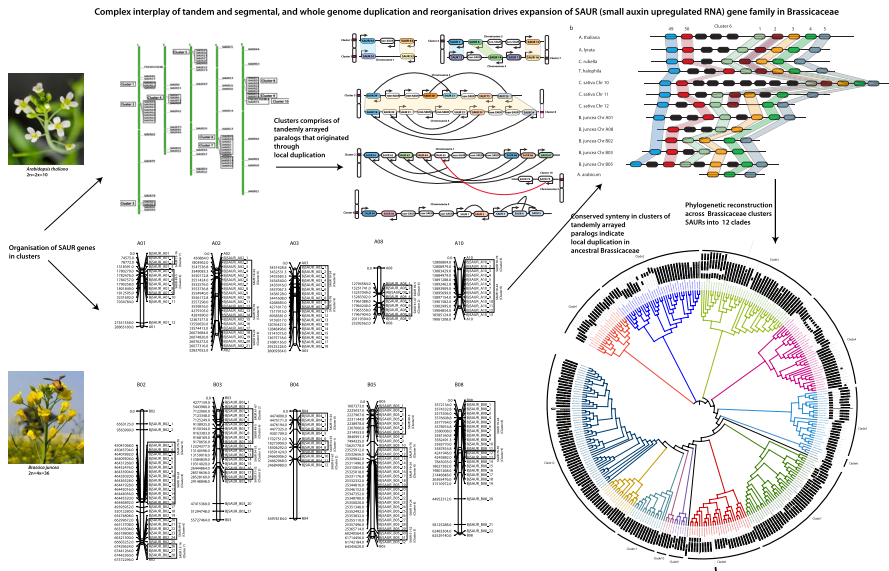
Abstract

Members of the *SAUR*, small auxin upregulated RNA, gene family initially identified as auxin inducible, mediate diverse developmental and adaptive processes in plants. In spite of their importance, identification and analysis of homologs from *Brassica juncea*, a major oilseed crop, is lacking. Additionally, investigations into organisational complexity and evolutionary past across Brassicaceae remain to be investigated. The present study was therefore designed to identify members of the *SAUR* gene family in *B. juncea*, reconstruct phylogenetic relationship, and analyse the history of expansion of the *SAUR* gene family across Brassicaceae. Genome-wide *in-silico* analysis allowed us to identify 237 *SAUR* genes in the allotetraploid *B. juncea* (AABB genome), which are distributed in a clustered manner among all 18 chromosomes of the *B. juncea* genome. Comparative analysis with the diploid parents- *B. rapa* (AA) and *B. nigra* (BB) revealed conserved organisation pattern. A striking feature of *SAUR* genes is intronless nature of most members. Comparative analysis revealed ten clusters of tandemly arrayed genes (TAGs) in *Arabidopsis thaliana*; two of these clusters were lost, and 33 clusters that are orthologous to the rest of *A. thaliana* clusters were identified from *B. juncea* genome. Organisational complexity revealed the presence of putative bidirectional promoters between some *SAUR* genes. Phylogenetic reconstruction shows several *SAUR* genes of *A. thaliana* and *B. juncea* forming separate clades, indicating lineage-specific expansion. Inclusion of homologs from across Brassicaceae allowed us to perform comparative synteny analysis and hypothesize local duplications being responsible for the tandem organisation, and segmental duplications as driving mechanism for large-scale expansion. The present study allowed us to catalog homologs of the *SAUR* gene family in *B. juncea*. This study thus forms the foundation for functional characterization involving transcriptional regulation, generation, and analysis of reverse genetic models toward understanding their role in plant growth and development.

✉ Sandip Das
sdas@botany.du.ac.in; sandipdas04@gmail.com

¹ Department of Botany, University of Delhi, Delhi 110007, India

Graphical Abstract



Keywords SAUR · Tandemly arrayed genes (TAG) · *Brassica* · Synteny · Local tandem duplication · Brassicaceae

Introduction

Auxin regulates multiple aspects of plant growth and development (Teale et al. 2006) such as floral organ and fruit development (Serrani et al. 2008; Zhang et al. 2023), apical dominance (Cline et al. 1997), phototropism (Bruinsma and Hasegawa 1990), cell proliferation and elongation (Woodward and Bartel 2005), root initiation, and response to various external and internal stimuli. Auxin arbitrates these processes at the molecular level through three classes of auxin-inducible gene families, *AUX/IAA*, *Gretchen Hagen3 (GH3)*, and *small auxin up-regulated RNA (SAUR)*. Among these three, the *SAUR* gene family has been reported to be the largest (Ren and Gray 2015). *SAUR* genes were first reported in elongating soybean hypocotyl and up-regulated in response to auxin (McClure and Guilfoyle 1987).

SAUR homologs have been identified in several plant species. These include *Physcomitrella patens* (18; Zhang et al. 2021), *A. thaliana* (79 + 2 pseudo-genes; Markakis et al. 2013), *Sorghum vulgare* (71; Wang et al. 2010), *S. bicolor* (71; Wang et al. 2010), *Citrullus lanatus* (65; Zhang et al. 2017), *Cucumis sativus* (73; Luan et al. 2023), *Oryza sativa* (60; Jia et al. 2024; and 58; Jain et al. 2006), *Dimocarpus longan* (68; Chen et al. 2023a, b, c), *Setaria italica* (72; Ma et al. 2023), *Prunus avium* (86; Hou et al. 2023) *Santalum album* (43; Zhu et al. 2024), *Glycine max* (236; Zhang et al. 2021), *Solanum lycopersicum* and *S. tuberosum* (99, and

134, respectively; Wu et al. 2012), *Zea mays* (79; Chen et al. 2014), *Gossypium raimondii*, *G. arboreum*, *G. hirsutum*, and *G. barbadense* (145, 97, 214, and 176, respectively; Li et al. 2017) and, *Populus trichocarpa* (105; Hu et al. 2018). In a recent analysis, 795 SAUR protein sequences derived from major plant clades represented by *Anthoceros*, *Marchantia*, *Physcomitrella*, *Selaginella*, *Ginkgo*, *Gnetum*, *Amborella*, *Arabidopsis*, *Oryza*, *Zea*, *Glycine*, *Medicago*, and *Setaria* have been used to perform phylogenetic reconstruction (Zhang et al. 2021). Several of these studies have reported the preponderance of intronless nature and clustered organisation (Tian et al. 2022; Zhang et al. 2017; Chen et al. 2014; Li et al. 2017). Additionally, studies have also speculated on the role of tandem and segmental duplication events in the evolutionary history of the SAUR family (Liu et al. 2022; Gan et al. 2020; Li et al. 2017).

SAUR proteins consist of a conserved domain of around 60 residues with low similarity at the N- and C-termini (Jain et al. 2006; Park et al. 2007; Ren and Gray 2015). The conserved signature SAUR domain may be responsible for the core function; whereas variance at the N- and C-terminal may be responsible for roles in diverse developmental and adaptive processes. For instance, N- and C-terminal of some SAUR proteins in *A. thaliana*, tomato, potato, and sorghum were reported to have metal binding capacity (Wu et al. 2012), and the N-terminal of some SAURs in *A. thaliana*, maize, and soybean bind to calmodulin (Yang and Poovaiah 2000; Popescu et al. 2007). The presence of SAUR-specific downstream element (DST element) was also reported in 30 out of 79 SAUR genes in *A. thaliana* (Ren and Gray 2015) and other plant species (Wu et al. 2012; Jain et al. 2006, and were found to regulate half-life and decay of SAUR mRNA (Johnson et al. 2000).

SAUR proteins have been shown to be involved in the elongation process by regulating the H⁺ ATPase pump and inhibiting PP2CD phosphatases. (Chae et al. 2012; Spartz et al. 2012, 2014; Stamm and Kumar 2013). In *A. thaliana*, overexpression of SAUR32 negatively affects hypocotyl growth and apical hook formation (Park et al. 2007). In contrast, in cucumber, overexpression of SAUR31 leads to longer roots and hypocotyl length (Luan et al. 2023).

Even though SAUR genes were discovered as being auxin induced, expression pattern of different members of SAUR genes was found to be regulated by Gibberellic acid (Bai et al. 2012), Jasmonic acid (Nemhauser et al. 2006), Brassinosteroid (Oh et al. 2014), Cytokinin (van Mourik et al. 2017), Ethylene (Shin et al. 2019), Absciscic acid (Kodaira et al. 2011) and environmental factors such as light (Ren and Gray 2015), drought, cold, salinity (Wu et al. 2012; Zhang et al. 2023), and high temperature (Franklin et al. 2011). A group of 32 redundantly acting SAUR genes were reported to be involved in light-regulated seedling growth in *A. thaliana* (Spartz et al. 2014), and a complex interaction between ARF6 (auxin), BZR1 (Brassinosteroids), and PIF1 (light) regulates the downstream expression of multiple SAUR genes in hypocotyl (Oh et al. 2014). TCP and PIF have also been reported to induce the expression of SAUR16 and SAUR15 via the ARF-BZR pathway (Danisman et al. 2012; Challa et al. 2016; Dong et al. 2019).

Members of Brassicaceae harbour morphological, ecological, and genetic diversity and are also valued for their economic and scientific importance. The complex evolutionary history of Brassicaceae is marked by multiple whole genome

duplications (α , β , and γ events; Kagale et al. 2014; Tiley et al. 2016; Qiao et al. 2019) and allopolyploidization events (e.g. origin of *B. juncea*, *B. napus*, *B. carinata*, *C. sativa*; Nagaharu 1935; Mandáková et al. 2019) making it an excellent model family to explore evolutionary processes related to gene families (Franzke et al. 2011; Kang et al. 2021). The availability of complete genome sequences of a number of Brassicaceae species and their close evolutionary relationships with *A. thaliana* also allows us to perform comparative genomic analysis to understand evolutionary trajectories, as has been performed for members of *MYB* (Lal et al. 2022; Yadav et al. 2023), *OVATE / OFP* (Chahar et al. 2021), *MIR159* (Anand et al. 2019), and *MIR319* (Joshi et al. 2018) gene families.

In spite of advances made in the identification of *SAUR* family members, several glaring lacunae remain especially with respect to complete cataloguing and *in-silico* analysis of properties across members of Brassicaceae. This study was therefore designed with specific goals to identify, prepare a compendium of *SAUR* genes, analyse genome-level distribution patterns, and perform comparative genomics of tandemly arrayed gene clusters to uncover the evolutionary past. The present study, to the best of our knowledge, is the first comprehensive survey to identify and perform *in-silico* analysis of homologs of *SAUR* genes across selected members of Brassicaceae. The results of this study provide an insight into the complex evolutionary pattern involving a combination of local tandem and segmental duplication, and form a framework for functional characterization in future.

Methods

Identification and Confirmation of *SAUR* Homologs

The CDS and protein sequences of 79 *SAUR* genes and 2 pseudo-genes of *A. thaliana* identified earlier were downloaded from the TAIR database using “*SAUR*” as a keyword and using previously annotated gene IDs (<http://www.arabidopsis.org/>; Supplementary Table 1; Hagen and Guilfoyle 2002). *A. thaliana* *SAUR* CDS sequences were used as a query for BLASTN against *B. juncea* var. *tumida* genome v.2.0 (Yang et al. 2016) and CDS sequences of *B. rapa* (Brara Z1.v2; Istace et al. 2021) and *B. nigra* (Bnigra NI100.v2; Perumal et al. 2020) using default parameters (e-value = 10^{-3} ; num alignments = 100). High-scoring pairs (HSP) were downloaded from the BRAD using the BLAST feature of TBtools-II software (Chen et al. 2023a, b, c), sorted in MS-Excel and manually examined to remove redundant sequences.

To identify tandemly arranged *SAUR* gene clusters in other Brassicaceae members, genomes of *Arabidopsis lyrata*, *Capsella rubella*, *Camellina sativa* and *Thellungiella halophila* were downloaded from the BRAD (BRAD v.3; <http://brassicadb.cn>), and that of *Aethionema arabicum* from *A. arabicum* DB (https://plantcode.cup.uni-freiburg.de/aetar_db/downloads.php). Orthologs of *SAUR* gene clusters were identified by performing a local BLAST (parameters: e-value = 10^{-3} ; num alignments = 100) using tandemly arranged *SAUR* genes of *A. thaliana* as query. Gene coordinates were identified by mapping predicted *SAUR* sequences with their respective genomes.

Predicted SAUR sequences were confirmed through Interpro and CDD batch search (<https://www.ncbi.nlm.nih.gov/Structure/bwrpsb/bwrpsb.cgi>) for the presence of the conserved SAUR domain (InterPro entry- GO:0009733, pfam- PF02519, panther- PTHR31374; Paysan-Lafosse et al. 2023). Only those sequences that contain the SAUR domain were considered for further analysis. Motif analysis of predicted *B. juncea* SAUR proteins was performed using MEME Suite v.5.5.5 (Multiple Em for motif Elicitation; <https://meme-suite.org/meme/>; Bailey et al. 2015) with the following parameters: motif discovery mode=classic; Number of motifs=15; Site distribution=zero to one occurrence per sequence; and, motif width=6 to 50. The output file was downloaded in XML format for visualization. The gene and CDS sequences of *B. juncea* SAUR genes were used to predict and analyze Exon–Intron structure using GSDS v.2 (Gene structure display server; <http://gsds.gao-lab.org/>; Hu et al. 2015).

Orthology and Synteny Analysis

Orthology of the putative *SAUR* genes was established using a 100 kb region comprising candidate *SAUR* gene/s flanked by 50 kb upstream and 50 kb downstream sequence through AVID and gVISTA (<https://genome.lbl.gov/cgi-bin/GenomeVista>; Bray et al. 2003) based global alignment against the *A. thaliana* genome and followed by visualization following previously established pipeline (Anand et al. 2019; Bhardwaj et al. 2020; Chahar et al. 2021; Lal et al. 2022). Orthologous clusters with more than 80% alignment with the *A. thaliana* genome were considered true and were taken for further analysis. For comparative genomic and synteny analysis, the ca. 100-kb region was used for *ab initio* gene prediction employing FGENESH (Solovyev et al. 2006), followed by Blast2GO annotation of the genes using Omicsbox v 3.0.27 (<http://www.biobam.com/omicsbox>) as described (Lal et al. 2022; Yadav et al. 2023). Annotated *SAUR* gene sequences were retrieved from the FGENESH file for further analysis.

Chromosomal Distribution and Organisation of SAUR Genes

Chromosomal lengths of *A. thaliana*, *B. juncea* and *B. rapa* were retrieved from TAIR (https://www.arabidopsis.org/portals/genAnnotation/gene_structural_annotation/agicomplete.jsp) and Ensemble plants (<https://plants.ensembl.org/index.html>), and of *B. nigra* from NCBI (https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_016432835.1/), respectively. The coordinates of predicted orthologs were obtained from the results of BLASTN performed against the respective genomes. Genes were mapped on chromosomes according to their gene coordinates and were visualized by MapChart 2.32 (Voorrips 2002). Bidirectional and reverse BLAST analysis was performed using SAUR proteins of *B. juncea* to predict duplicated gene pairs and results obtained after intra-genomic BLASTN with > 80% similarity were

considered paralogs. SAUR genes obtained within a 50 kb region were regarded as one cluster.

Gene Duplication and Synteny Analysis

Gene duplication events were defined according to parameters of Li et al. (2017), which states that the alignment should cover more than 80% of the longer gene with at least 80% identity in the aligned region (Li et al. 2017). The genes located in a window of 50 kb region were considered to be tandemly arranged; if paralogs were present outside this 50 kb region, they were supposed to be segmentally duplicated. Duplications were visualized using shinyCircos tool (Wang et al. 2023). shinyCircos-based diagrams were built for *B. juncea* (separately for A and B genomes) to identify segmental and tandem duplication as a mode of origin. Visual representations of gene clusters were shown by manually constructing microsynteny maps on pseudochromosomes using Adobe Illustrator 2024 (Adobe; USA).

Tandemly and segmentally duplicated pairs of *B. juncea* SAUR members were used to calculate non-synonymous substitution rate (Ka) and synonymous substitution rate (Ks) to calculate Ka/Ks (ω) ratio. Homologous protein sequence pairs (protein1 / protein 2) were compiled as a tab-delimited file as input to compute the Ka, Ks and Ka/Ks ratio using TBtools-II (Chen et al. 2023a, b, c).

Physicochemical Properties

ProtParam ExPasy server (<https://web.expasy.org/protparam/>; Gasteiger et al. 2005) with default parameters was used to determine the physicochemical profiling of SAUR proteins, including molecular weight (MW), GRAVY value, pI value, instability index, and aliphatic index; sub-cellular localization of proteins was predicted using WoLF PSORT (<https://wolfpsort.hgc.jp>) (Horton et al. 2007) as described earlier (Lal et al. 2022; Chahar et al. 2021).

Phylogenetic Analysis

Phylogenetic reconstruction across homologs of SAUR from *A. thaliana* and *B. juncea* was done using full-length protein sequences of *A. thaliana* SAUR members and predicted homologs from *B. juncea*. Multiple sequence alignment was performed using MUSCLE (seqtype=protein; gap open=-2.9; gap extend=0.0; cluster method=UPGMA; Min Diag length (lambda)=24) implemented in MEGA-X software (Edgar 2004; Kumar et al. 2018), and the aligned file was saved as.mas /.masx format. A maximum likelihood tree was constructed using the aligned file keeping default parameters with 1000 bootstrap value, Substitution model=Jones–Taylor–Thornton (JTT) (test of phylogeny=none; model=Jones–Taylor–Thornton (JTT); rates among sites=uniform; gaps/missing data=use all sites; ML Heuristic method=Nearest Neighbour Interchange; Initial

Tree for ML=Make initial tree automatically (NJ/MP); branch swap filter=none; number of threads=4).

Phylogeny among SAUR homologs of all tandemly arranged homologs across Brassicaceae was reconstructed based on protein sequences employing a Bayesian tool (BEAST; <https://github.com/beast-dev/beast-mcmc/releases>; Suchard et al. 2018; Drummond et al. 2002; Pybus and Rambaut 2002; Rambaut 2000) using previously described steps (Lal et al. 2022; Chahar et al. 2021). Briefly, multiple sequence alignment of protein sequences was done using MUSCLE in MEGA-X (Edgar 2004) as described above, with masking to remove residues present in less than 3% sequences. The alignment output was saved as NEX file format which was used as input for BEAUti to generate.xml file (parameters: Length of chain=10,000,000 generations; substitution model=Blosum62; Site Heterogeneity model=none; no. of Gamma categories=4; Strict clock; Tree=Coalescent, constant size). The.xml file was then used as input for Bayesian analysis (BEAST v1.8.4) to generate a.tree file, which was then annotated using TreeAnnotator v1.8.4 (burnin state=0; posterior probability limit=0; maximum clade credibility tree; tree cut-off as 10,000; Drummond et al. 2012; Bouckaert et al. 2014). Trees were visualized and annotated using FigTree (<https://github.com/rambaut/figtree/releases>; Rambaut 2018). All trees were edited using the iTOL online tool (<https://itol.embl.de/>).

Protein–Protein Interaction Map

Protein–protein interaction network of *B. juncea* SAUR proteins was constructed and analyzed by STRING v12 online tool (<https://string-db.org/>) using *A. thaliana* as a reference.

GO Analysis

SAUR protein sequences (in FASTA format) were used as queries in Blast2GO, a functional analysis module of Omicsbox v 3.0.27 (<https://www.biobam.com/omicsbox/>), and analysed using default parameters.

Promoter Analysis

A 1500 bp region upstream of the translational start site of *B. juncea* SAUR genes were manually retrieved. Cis-elements related to Transcription factor binding sites (TFBS) were predicted using the PlantCare (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) tool and visualized using TBtools-II as described (Chen et al. 2023a, b, c).

Fig. 1 Chromosomal locations and paralogous relationship of clusters. **A** Mapping of *SAUR* genes of *Arabidopsis thaliana* on the chromosomes based on genomic coordinates using “Chromosome Map” tool of TAIR (www.arabidopsis.org). The clusters have been boxed and indicated. **B** Diagrammatic representation of chromosomal location (red box on chromosome) and paralogy among and between members of *SAUR* clusters. Paralogous clusters were identified based on phylogenetic reconstruction and joined using either shaded boxes or curved lines. Transcriptional orientation is marked by bent arrows

Results

SAUR Genes Constitute a Large Family in *Brassica* Species

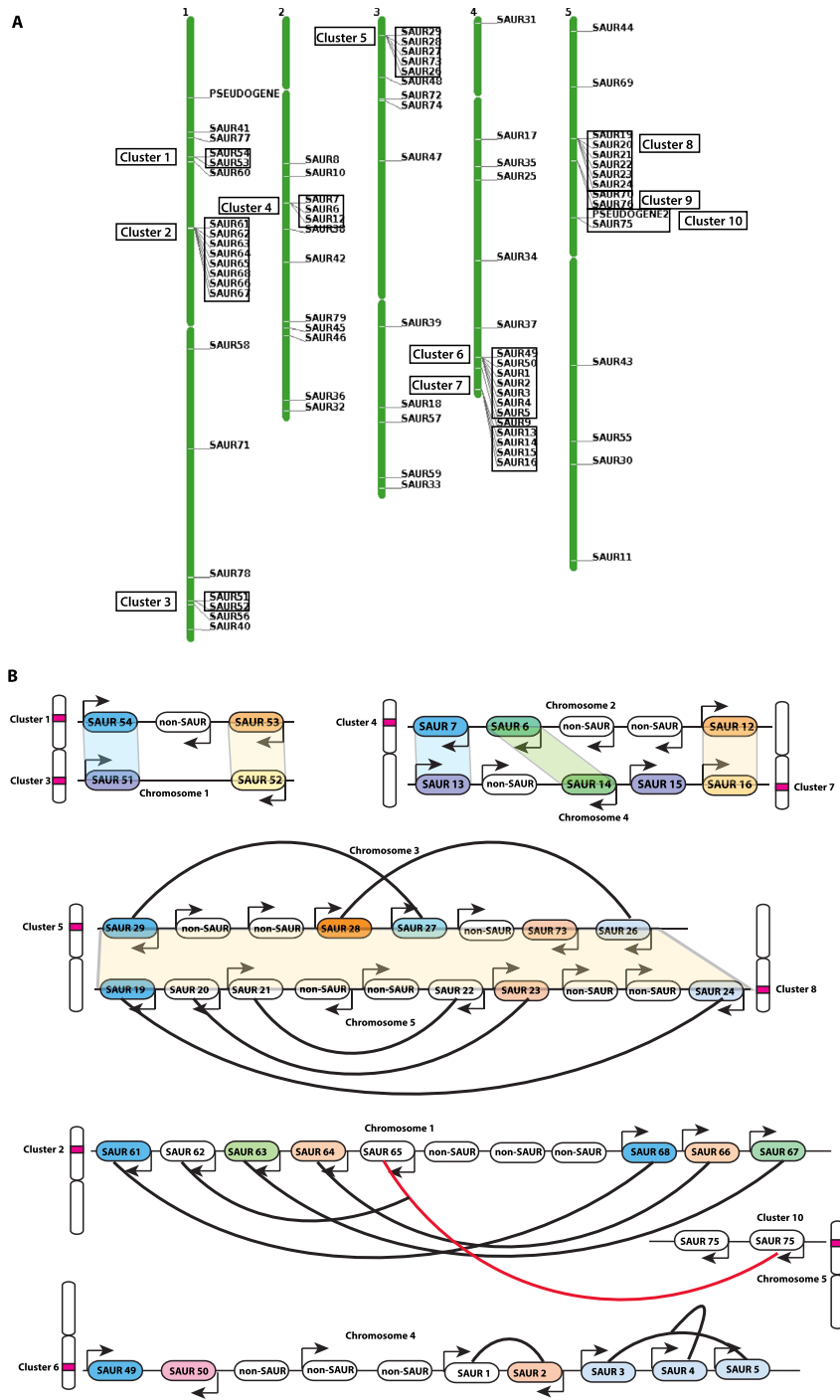
A total of 237 homologs were identified from *B. juncea* genome (AABB). A comparative analysis revealed 99 members in *B. rapa* (AA) and 107 members in *B. nigra* (BB), the two parental genomes. The gene length varied between 267 to 1038 bp in *B. rapa*, 222 to 1200 bp in *B. nigra*, and 225–1014 bp in *B. juncea* indicating nearly conserved gene length. The predicted mean length was slightly longer in *B. rapa* (403 bp) as compared to *B. nigra* (384 bp) and *B. juncea* (359 bp) (Supplementary Table 1; Supplementary Fig. 1).

NCBI-CDD batch search and Pfam database and visualization of results using TBtools-II confirm the presence of a conserved core of auxin-inducible SAUR domain in all the homologs (Supplementary Tables 2.1, 2.2).

Accurate identification and unique nomenclature are essential for the analysis of *SAUR* genes. We first attempted to annotate *Brassica SAURs* based on their orthology to *A. thaliana SAUR* genes. Owing to extremely high levels of sequence similarity at both CDS and protein levels, *SAUR* genes could not be unambiguously assigned to their respective orthologs. Annotation was therefore done first according to their chromosomal number, starting from chromosome A01 to A10, and B01 to B08, and location within each chromosome, starting from the top (p-arm) to the bottom (q-arm). For instance, A01_10 gene is present on the A01 chromosome and is 10th starting from the top arm. Homologs of *AtSAUR18*, *AtSAUR6*, *AtSAUR2*, *AtSAUR30*, *AtSAUR36*, *AtSAUR58*, *AtSAUR48*, *AtSAUR71*, and *AtSAUR40* could not be detected in *B. juncea* and also in the two parental genomes (*B. rapa* and *B. nigra*) based on high scoring pair results.

Chromosomal Distribution Reveals Clustered Organisation with Conservation and Divergence Across Species

A comparative analysis of distribution of *SAUR* genes on *A. thaliana*, *B. juncea*, *B. rapa* and *B. nigra* was performed (2s. 1, 2; Supplementary Tables 3.1; 3.2; Supplementary Fig. 1). *A. thaliana SAUR* genes were found to be distributed among all five chromosomes. Chromosome 1 has the maximum number of 20 *SAUR* genes and one pseudogene (21), whereas chromosome 2 has the least (12). In *B. juncea*, of the 237 *SAUR* genes, 106 were located on ten A-chromosomes and 130 genes were on eight B-chromosomes; one gene was mapped on a contig. Some of the genes were dispersed on chromosomes, whereas the majority formed clusters such as on A01, A02, A03, A08, and A10 in A-genome, and B02, B03, B04, B05, and B08 in B-genome.



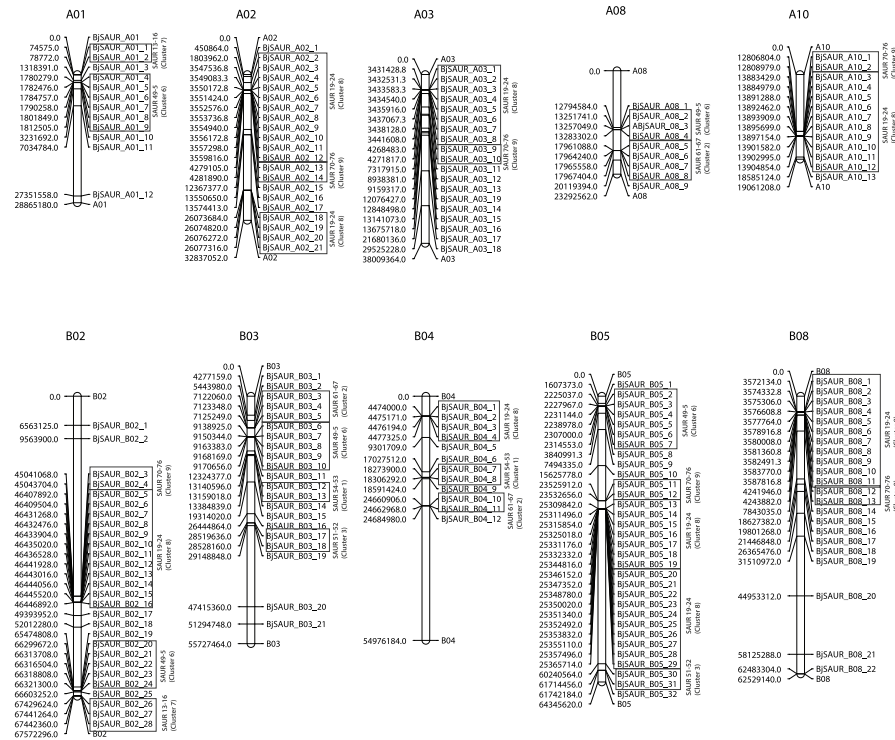


Fig. 2 Representative images of mapping and cluster organisation of *Brassica juncea* SAUR homologs on pseudochromosomes. Clusters are identified, boxed, and numbered based on phylogenetic reconstruction-based clustering of the *B. juncea* SAURs to *A. thaliana* homologs. Note the conserved organisation of clusters 6–7 and 8–9 in both *A. thaliana* and *B. juncea*

In A-genome, the maximum and minimum numbers were on A02 (19) and A04, A05 (4 each); in B-genome, maximum number of the SAUR genes was detected on chromosome B05 (32), and minimum on chromosome B07 (3) (Fig. 2).

It is critical to understand chromosomal organisation to understand the evolution and expansion patterns of SAUR genes at a genomic level. A detailed analysis of chromosomal organisation revealed that 39 out of 81 SAUR genes in *A. thaliana* are organised in ten clusters. We numbered these as cluster 1 to 10 based on their location, starting from the top arm of chromosome 1 (cluster 1) to the bottom arm of chromosome 5 (cluster 10). On closer inspection, these ten clusters could be sub-classified as being undisrupted (clusters 3, 4, 9) or disrupted (clusters 1, 2, 5, 6, 7, 8, 10) (Table 1). Undisrupted clusters were defined as two SAUR genes present adjacent to each other, within a distance of <5 kb, without any other intervening protein-coding gene/s, whereas disrupted clusters harboured non-SAUR genes between two SAUR genes.

These clusters were detected on all five chromosomes of *A. thaliana* with chromosome 1 and 5 having three cluster each (1–3; 8–10, respectively), chromosome 2 and 3 having one each (4, and 5, respectively), chromosome 4 having two

Table 1 Tandemly arrayed SAUR gene cluster in *A. thaliana*, *B. juncea*, *B. rapa* and *B.nigra*

<i>Arabidopsis thaliana</i>	SAUR genes	Homologous clusters in <i>Bras-sica juncea</i>	Homologous clusters in <i>Bras-sica rapa</i>	Homologous clusters in <i>Bras-sica nigra</i>
Genes, cluster nomenclature and chromosome				
Cluster 1 Chromosome 1	SAUR 54-53	Bj-B03-Clus1 Bj-B04-Clus1	BraA02- Clus1 BraA06 Clus1	BniB03 Clus1 BniB04 Clus1
Cluster 2 Chromosome 1	SAUR 61-62-63- 64- 65- 68- 66-67	Bj-A08-Clus2 Bj-A09-Clus2 Bj-B03-Clus2 Bj-B04-Clus2	BraA08 Clus2 BraA09 Clus2	BniB03 Clus2 BniB04 Clus2
Cluster 3 Chromosome 1	SAUR 51-52	Bj-A07-Clus3 Bj-B03-Clus3 Bj-B05-Clus3	BraA02 Clus3 BraA07 Clus3	BniB03 Clus3 BniB05 Clus3
Cluster 4 Chromosome 2	SAUR 7-6-12	No homologous cluster	No homologous cluster	No homologous cluster
Cluster 5 Chromosome 3	SAUR 29-28-27-73-26	No homologous cluster	No homologous cluster	No homologous cluster
Cluster 6 Chromosome 4	SAUR 49-50-1-2-3-4-5	Bj-A01-Clus6 Bj-A08-Clus6 Bj-B02-Clus6 Bj-B03-Clus6 Bj-B05-Clus6	BraA01 Clus6 BraA03 Clus6 BraA08 Clus6	BniB02Clus6 BniB03 Clus6 BniB05 Clus6
Cluster 7 Chromosome 4	SAUR 13-14-15-16	Bj-A01-Clus7 Bj-A05-Clus7 Bj-B02-Clus7	BraA01 Clus7 BraA01 Clus7	BniB02 Clus7 BniB05 Clus7

Table 1 (continued)

<i>Arabidopsis thaliana</i>	<i>SAUR</i> genes	Homologous clusters in <i>Bras-sica juncea</i>	Homologous clusters in <i>Bras-sica rapa</i>	Homologous clusters in <i>Bras-sica nigra</i>
Cluster 8 Chromosome 5	<i>SAUR 19-20-21-22-23-24</i>	Bj-A02A-Clus8 Bj-A02B-Clus8 Bj-A03-Clus8 Bj-A10-Clus8 Bj-B02-Clus8 Bj-B04-Clus8 Bj-B05-Clus8 Bj-B08-Clus8	BraA02A Clus8 BraA02B Clus8 BraA03 Clus8 BraA10 Clus8	BniB02 Clus8 BniB04 Clus8 BniB05 Clus8 BniB08 Clus8
Cluster 9 Chromosome 5	<i>SAUR 70–76</i>	Bj-A02-Clus9 Bj-A03-Clus9 Bj-A10-Clus9 Bj-B02-Clus9 Bj-B08-Clus9	BraA02 Clus 9 BraA03 Clus 9 BraA10 Clus 9	BniB02 Clus 9 BniB05 Clus 9 BniB08 Clus 9
Cluster 10 Chromosome 5	<i>Pseudogene-SAUR75</i>	Not analysed	Not analysed	Not analysed

clusters (6–7) (Fig. 1A). Composition revealed that each cluster was of variable size with cluster 2 (8 *SAUR* genes), cluster 6 (7 *SAUR* genes), and cluster 8 (6 *SAUR* genes) among the largest; clusters 1, 3, 9, and 10 were smallest with two *SAUR* genes each (Table 1). As cluster 10 contains one pseudogene and *SAUR* 75, homologs of this cluster were not identified for any further analysis.

As mentioned previously, several of the clusters harboured non-*SAUR* genes located in close proximity and these included cluster 1 (one non-*SAUR* gene), cluster 5 and 7 (two non-*SAUR* genes each), cluster 2 and 6 (three non-*SAUR* genes each), and cluster 8 (four non-*SAUR* genes arranged in two pairs). These all were separated from *SAUR* genes by < 2 kb and were therefore considered part of a single genomic block for comparative synteny analysis.

Analysis of the chromosomal location on the *B. juncea* pseudochromosome revealed a total of 32 clusters. These were found to be homologous to six out of nine clusters of *A. thaliana*. Homologs of *SAUR* 7-6-12 (cluster 4) and *SAUR* 29–26 (cluster 5) were not detected in the *B. juncea* genome. Among the seven *A. thaliana* clusters, the maximum homologs and homeologs were detected for *SAUR* 19–24 (cluster 8; Table 1), and the least number (two, on B03 and B04) for *SAUR* 54–53 (cluster 3; Table 1).

SAUR genes of *B. rapa* and *B. nigra* were mapped on pseudochromosomes and were found to be present on all chromosomes- A01-10 and B01-B08, respectively, and visualized using Mapchart (Supplementary Table 3.2; Supplementary Fig. 1). *SAUR* homologs identified in *B. rapa* and *B. nigra* were analysed for conservation with nine clusters of *B. juncea* (Table 1; Supplementary Table 4). Out of nine clusters, seven clusters were detected in the genomes of *B. rapa* and *B. nigra*, albeit with deletions and duplications. These include such examples as cluster 1, where tandemly organised genes were observed on the B03 and B04 chromosomes of *B. juncea*. In contrast, a single copy of the gene was observed in *B. nigra* B03 and B04. In contrast, homologs present on A02 and A06 chromosomes of *B. rapa* were found to be deleted from *B. juncea*. Cluster 2 is conserved in *B. juncea* and its progenitor genomes, except that a deletion was observed on *B. juncea* A09. In cluster 3, species-specific tandem duplication was observed on B03 and B05 of *B. juncea*. In cluster 6, species-specific tandem duplications were observed on A01, B03, and B02. The maximum number of species-specific duplications was observed in cluster 8 of *B. juncea* on A02, A03, A10, B02, B04, B05, and B08. In cluster 9, duplication was observed on A03, A10, and B08 in *B. juncea*, whereas single copies were found in *B. rapa* (A03 and A10) and *B. nigra* (B08) genomes (Table 1; Supplementary Table 4).

In the previous section, we described that the *SAUR* genes in *A. thaliana* and *B. juncea* are organised as clusters. We therefore generated separate phylogenetic trees using *SAUR* proteins from *A. thaliana* (Supplementary Fig. 2) and *B. juncea* (Supplementary Fig. 3) to understand the relationship between organisation pattern and phylogenetic clustering. The members of a given cluster in *A. thaliana* were generally found to group together; for example, the members of *SAUR* 61–68 (Cluster 2) formed a distinct clade, albeit with *SAUR*75 (from chromosome 5). An analysis of the *SAUR* genes in each cluster and their clade-wise arrangement revealed that the *SAUR* 54–53 cluster was paralogous to clusters of *SAUR*

51–52, with both clusters being on chromosome 1. Similarly, members of the cluster comprising *SAUR* 7–6–12 (on chromosome 3) were paralogous to *SAUR* 13–14–16 (cluster 7; chromosome 4). Members of cluster 2 on chromosome 1 and cluster 6 on chromosome 4 showed within-cluster paralogy; members of cluster 5 on chromosome 3 and cluster 8 on chromosome 5 also showed within-cluster paralogy, and the two clusters themselves formed sister clades, indicating their paralogous nature (Fig. 1B).

Mapping of the *B. juncea* *SAUR* homologs from 32 clusters of homologs and homeologs on the pseudo-chromosomes provided a global overview of organisation and evolutionary pattern. Analysis of the phylogenetic tree revealed that homologs from most clusters are grouped together irrespective of whether these are located on the A- or B-genome. Within each of these clades, homologs from A and B genomes grouped together. For instance, we identified seven genomic loci of *B. juncea* that group with *A. thaliana* *SAUR* 19–24 cluster and were considered orthologous; and six loci orthologous to *A. thaliana* *SAUR* 70–76 cluster; each genomic loci harbours multiple *SAUR* genes. The seven loci orthologous to *SAUR* 19–24 cluster (6 genes) are annotated as *BjA02-1* (10 *SAUR* homologs), *BjA02-2* (4), *BjA03* (8), *BjA10* (10), *BjB02* (12), *BjB04* (4), *BjB05* (17) and *BjB08* (11 homologs) and contain a total of 76 *SAUR* genes. All these 76 *BjSAURs* that are homologous to the members of *A. thaliana* *SAUR* 19–24 form a single clade (Fig. 2; Supplementary Fig. 3). Similarly, the six loci orthologous to *A. thaliana* *SAUR* 70–76 cluster (2 *SAUR* genes) are *BjA02*, *BjA03*, *BjA10*, *BjB02*, *BjB05*, and *BjB08* with two *SAUR* genes each. All these 12 *SAUR* genes of *B. juncea* were also found to form a single clade (Fig. 2; Supplementary Fig. 3). In *A. thaliana*, the *SAUR* 19–24 clade (6 genes) and *SAUR* 70–76 clade (2 genes) are located adjacent to each other on chromosome 5. Our analysis of the *B. juncea* genome also revealed a similar organisation of the two *BjSAUR* 19–24 and *BjSAUR* 70–76 clusters on multiple genomic loci (Fig. 2; Supplementary Fig. 3). A similar organisation pattern was observed when the composite phylogenetic tree of *A. thaliana* and *B. juncea* was examined, and *SAUR* genes present in a cluster of *A. thaliana* and *B. juncea* were found to form a single clade (Fig. 3). Several *A. thaliana* *SAUR* homologs were found to be deleted from the *B. juncea* genome, whereas some were found to be duplicated. For example, homologs of *A. thaliana* *SAUR* 7–6–12 (chromosome 2; cluster 4) and *SAUR* 29–26 (chromosome 3; cluster 5) were found to be deleted from the *B. juncea* genome (Table 1).

A composite reconstruction of phylogeny using protein sequences from *B. juncea*, *B. rapa* and *B. nigra* based on Bayesian method revealed clustering of homologs based on their source genome affiliations (Supplementary Fig. 4). The *B. juncea* A-genome homologs grouped with homologs from *B. rapa*, and B-genome homologs with *B. nigra*, thus clearly establishing orthologous relationship. The clustering also allowed identification of homologs present in the parental genome but lost from *B. juncea* (e.g., *BnigraSAUR55*; *BnigraSAUR56*; *BrapaSAUR44*) or vice-versa (e.g., *BjunSAUR_B03_18*; *BjunSAUR_B04_8*) indicating species- or lineage-specific losses.

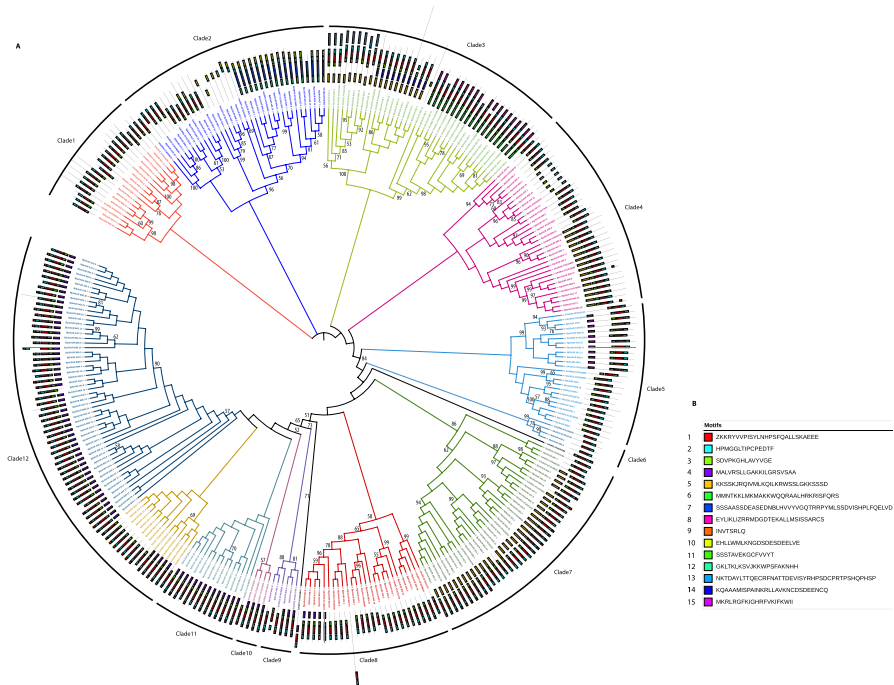


Fig. 3 **A** Phylogenetic reconstruction based on SAUR protein sequences reveals that homologs from *A. thaliana* and *B. juncea* from 12 distinct clades, and *Brassica*-specific expansion. **B** MEME analysis of SAUR proteins identified 15 conserved motifs which are marked on the SAUR proteins and depicted on the protein as radiating outer to the phylogenetic tree circle

Gene Structure Shows Predominant Intronless Nature and Small CDS of SAURs

Gene structure was analyzed to understand intron number, position, and length across the *B. juncea* SAUR gene family using GSDS v.2 software (<http://gsds.gao-lab.org/>). Results showed that out of 237 members 224 SAUR were intronless and 13 members had between 1 to 4 introns each. Of these 13, *BjSAUR_A02_13*, *BjSAUR_A06_2*, *BjSAUR_A10_2*, *BjSAUR_B02_4*, *BjSAUR_B02_8*, *BjSAUR_B05_12*, *BjSAUR_B05_16*, *BjSAUR_B07_1*, *BjSAUR_B08_20* and *BjSAUR_B08_22*, each have one intron, *BjSAUR_B08_12* has two introns, and *A03_9*, *B02_24* has four introns (Table 2; Supplementary Fig. 5).

The CDS length varied from 1014 nucleotides (nt) in *BjSAUR_A03_9* (337 amino acids; aa) to only 225 nucleotides in *BjSAUR_A04_4* (74 aa). No bias in average length distribution was observed across A-genome (average CDS across 106 SAUR = 360 nt; 119 aa) or B-genome (average CDS across 130 SAUR = 356 nt; 117 aa) copies. Frequency distribution revealed that nearly 74% of all SAUR CDS in both A-genome and B-genome fall in the 201–400 nucleotide range; the rest fall in the 401–600 nucleotide range (25.5% in A-genome and 24% in B-genome) (Supplementary Table 5).

Table 2 Representative sampling of SAUR family across plant lineages and analysis of intronless genes

Plant lineage	Family	Species	Total numbers	Percentage of intronless genes	References
Algae					
Bryophyte	Anthocerotaceae	<i>Anthoceros angustus</i>	3	Not available	Zhang et al. (2021)
	Marchantiaceae	<i>Marchantia polymorpha</i>	14	14* (100%)	Zhang et al. (2021)*
		<i>Physcomitrella patens</i>	18# / 9*	0	Rensing et al. (2008)#
Peridophyte	Selaginellaceae	<i>Selaginella moellendorffii</i>	15 # / 8*	6* (75%)	Zhang et al. (2021)
Gymnosperm	Ginkgoaceae	<i>Ginkgo biloba</i>	42	Not available	Zhang et al. (2021)
	Gnetaceae	<i>Gnetum montanum</i>	37	Not available	Zhang et al. (2021)
Angiosperm					
Basal angiosperm	Amborellaceae	<i>Amborella trichopoda</i>	25	24* (96%)	Zhang et al. (2021)*
Dicot	Solanaceae	Tomato	99	90 (90.9%)	Wu et al. (2012)
		Potato	134	131 (97.8%)	Wu et al. (2012)
	Malvaceae	<i>Gossypium raimondii</i>	145	124 (85.5%)	Li et al. (2017)
		<i>G. arboreum</i>	97	92 (94.8%)	Li et al. (2017)
		<i>G. hirsutum</i>	214	202 (94.4%)	Li et al. (2017)
		<i>G. barbadense</i>	176	155 (88.06%)	Li et al. (2017)
	Salicaceae	<i>Populus trichocarpa</i>	105	94 (89.5%)	Hu et al. (2018)
	Sapindaceae	<i>Dimocarpus longan</i>	68	52 (76.5%)	Chen et al. (2023a, b, c)
	Lamiaceae	<i>Salvia miltiorrhiza</i>	55	46 (83.6%)	Wang et al. (2024)
	Rosaceae	<i>Eriobotrya japonica</i>	95	85 (89.5%)	Peng et al. (2022)
		<i>Fragaria vesca</i>	66	38 (57.6%)	Kiyak et al. (2022)
	Cucurbitaceae	<i>Cucumis melo</i>	66	66 (100%)	Tian et al. (2022)
		<i>Benincasa hispida</i>	68	66 (7.05%)	Luo et al. (2022)
		<i>Citrullus lanatus</i>	65	65 (100%)	Zhang et al. (2017)
	Fabaceae	<i>Medicago truncatula</i>	133	123 (92.5%)	Zhang et al. (2021)

Table 2 (continued)

Plant lineage	Family	Species	Total numbers	Percentage of intronless genes	References
Monocot	Santalaceae	<i>Santalum album</i>	43	34 (79.06%)	Zhu et al. (2024)
	Brassicaceae	<i>Arabidopsis thaliana</i>	79 (excluding two pseudogenes)	73** (92.4%)	Zhang et al. (2021)
		<i>Brassica juncea</i>	237	224 (94.5%)	This study
	Rutaceae	<i>Citrus sinensis</i>	70 / 50*	39* (78%)	Zhang et al. (2021)
	Urticaceae	<i>Boehmeria nivea</i>	71	Not available	Zhang et al. (2021)
	Poaceae	<i>Setaria italica</i>	72	59 (81.9%)	Ma et al. (2023)
		<i>Oryza sativa</i>	58	58 (100%)	Jain et al. (2006)
		<i>Phyllostachys edulis</i>	38	32 (84.2%)	Bai et al. (2017)
		<i>Zea mays</i>	79	73 (92.4%)	Chen et al. (2014)

*Number of total and intronless SAURs retrieved from Phytozome13 or NCBI

***Number of total and intronless SAURs retrieved from TAIR

Number reported in publication

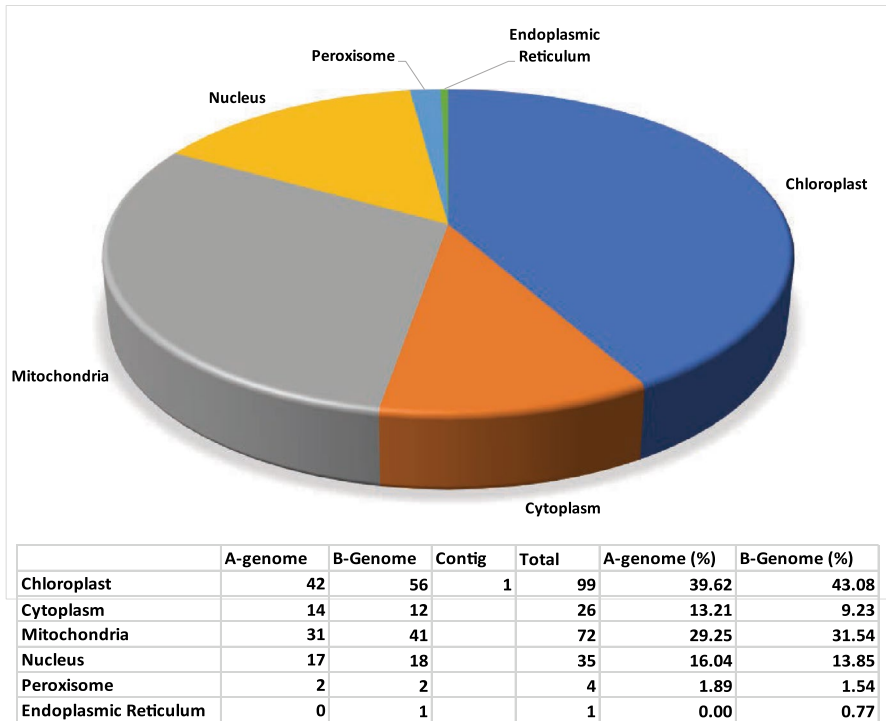


Fig. 4 Pie chart showing putative sub-cellular localization of 237 SAUR proteins from the A and B genome of *B.juncea* predicted using WoLF PSORT (<https://wolfsort.hgc.jp>). Note. Maximum number of SAUR genes are localised in the chloroplast followed by mitochondria

SAUR Proteins Exhibit Variable Physicochemical Properties and Motif Abundance

The predicted number of amino acids ranged from 74 (BjSAUR_A04_4) to 337 (BjSAUR_A10_5), predicted molecular weight ranged from 8438.85 Da (BjSAUR_B04_9) to 11,236.82 Da (BjSAUR_B02_12); theoretical isoelectric point (PI) ranged from 4.6 (BjSAUR_A04_4) to 11.36 (BjSAUR_A03_9); instability index varied from 16.94 (BjSAUR_B05_23) to 84.96 (BjSAUR_B03_1); aliphatic index ranged from 51.33 (BjSAUR_B03_1) to 109.19 (BjSAUR_A04_4); and the GRAVY value varied from 1.217 (BjSAUR_B03_1) to 0.286 (BjSAUR_A08_6) (Supplementary Table 6).

The sequences of BjSAUR proteins were analysed using MEME suite and 15 conserved amino-acid motifs were identified (motifs 1–15; Fig. 3; coloured boxes). Motif 2 was found to be the most abundant and was detected in 231 out of 237 SAUR proteins (except for A02_15, A04_4, A05_4, B01_4, B06_2, and B04_12), followed by motif 1, which was present in 218 members; motif 3 was present in 207 BjSAUR members.

Sub-cellular localisation analysis revealed that 26 out of 237 are cytoplasm localized, 99 in the chloroplast (maximum), followed by 72 in mitochondria, 34 in the nucleus; four were predicted to be located in peroxisomes, and one in the endoplasmic reticulum. The fraction of each category was nearly equal in both A- and B-genome derived SAURs thus not showing any genome bias (Fig. 4; Supplementary Table 6).

GO Analysis Confirms Auxin Responsiveness and Involvement in Signalling

GO annotation of *B. juncea* SAUR CDS was performed using Blast2GO, and the biological process revealed that most are auxin-responsive and involved in signal transduction. Cellular function revealed the presence of SAUR proteins in the plasma membrane, cell periphery, cytoplasm, nucleus, and peroxisomes; most were found to be cell membrane localized. Molecular processes reveal ion-binding, protein-binding, and metal-binding capacity of the SAUR proteins to perform downstream functioning, and were found to be involved in catalytic, hydrolase, endopeptidase, and peptidase activity (Supplementary Fig. 6; Supplementary Table 7).

Protein–Protein Interaction Reveals Both SAUR-SAUR, and, SAUR-non-SAUR Network

Potential protein–protein interacting network of *B. juncea* SAUR proteins was predicted using the STRING tool (<https://string-db.org/>) using *A. thaliana* protein data as a reference. A total of 77 pairs of SAUR protein-SAUR protein interactions were predicted within *B. juncea* family. Additionally, 132 interactors were also predicted that included non-SAUR proteins such as AHA (plasma membrane H(+) ATPase), AHL (AT-hook motif nuclear-localized protein), ASK (Protein kinase negatively regulates brassinosteroid signal transduction pathway) AUX-IAA (Auxin transporter protein-Auxin-responsive protein), BRI (Protein BRASSINOSTEROID INSENSITIVE), BAK (BRASSINOSTEROID INSENSITIVE 1-associated receptor kinase), BKI (BRI1 kinase inhibitor), BZR (Protein BRASSINAZOLE-RESISTANT), PP2C (Probable protein phosphatase), TCP (Transcription factor TCP), AUX1 (Auxin transporter protein 1), and ARF30 (Cupredoxin superfamily protein) (Supplementary Table 8; Fig. 5).

Promoter Analysis Unravels Diverse Cis-Regulatory Motif Landscape

For cis-regulatory element analysis, 1500 bp upstream region of the start site was considered as a putative promoter element, and cis-regulatory motifs were predicted by the PlantCare tool to identify stress-responsive elements across 237 SAUR promoters. Motifs include anaerobic response element (350) (ARE), MYBHv1 binding sites (46) (MybHv1), drought response elements (MYC), wound responsive elements (80), low-temperature response elements (82) (LTRs), and, anoxic-specific inducibility elements (10). Tissue-specific elements include zein metabolism

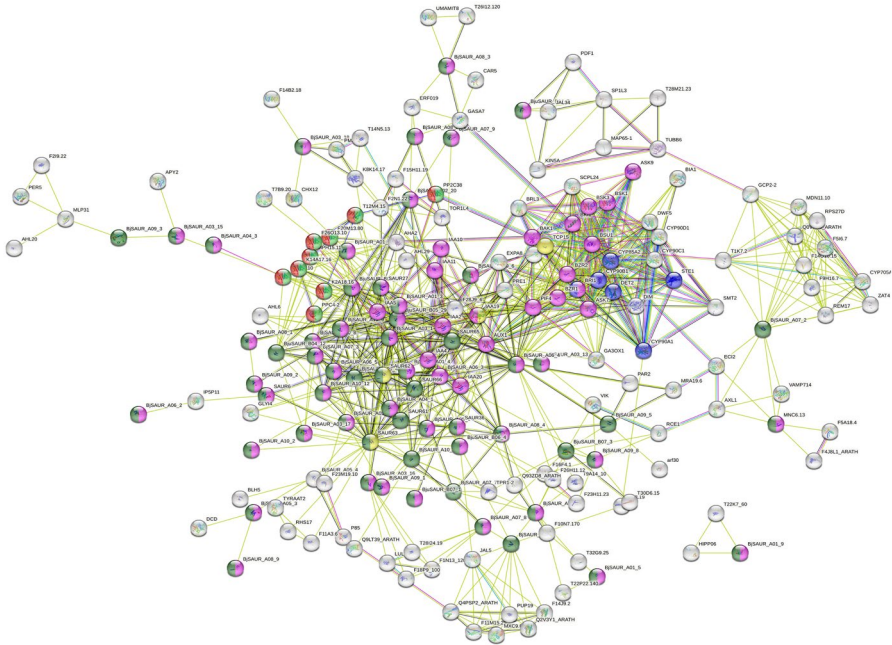


Fig. 5 Protein–protein interaction network of *B. juncea* SAUR proteins predicted using the STRING database (<https://string-db.org/>). Green spheres represent SAUR proteins, olive green lines represent the data obtained from text mining, blue lines indicate protein homology based, and pink line shows experimentally determined interactions

regulation elements (63), circadian control elements (39), meristem development elements (35), endosperm development elements (37), seed-specific elements (18), and flavonoid synthesis elements (14). In addition to these, several phytohormone-related elements were also observed like SARE (76; Salicylic acid response element), ABRE (276; abscisic acid response element), GRE (124; Gibberellic acid response element), MeJA (316; Methyl jasmonate) along with AuxRE (88; auxin response element). This revealed that the *BjSAUR* genes are not only regulated by auxin but also by other plant hormones. MYB_LRE sites were also found in the promoter region indicating their regulation by MYB transcription factors. In addition, cis-regulatory motifs associated with environmental factors such as light-responsive elements were also found indicating photo-regulation of members of *SAUR* gene family (Supplementary Table 9; Supplementary Fig. 7).

Phylogenetic Reconstruction, and Within- and Between-Cluster Paralogy Predicts Segmental and Local Duplication as a Mode of Evolution and Expansion

The expansion of the *SAUR* gene family vis-à-vis its genomic position has remained an enigma because of its clustered organisation and high sequence similarity. Phylogenetic analysis was therefore carried out between *A. thaliana* and *B. juncea*

SAUR proteins, between selected members of Brassicaceae, and via a synteny-based approach.

To understand the evolutionary relationship between *B. juncea* (237) and *A. thaliana* (79) SAUR gene family, an unrooted Maximum-Likelihood tree (1000 bootstrap) was constructed using full-length protein sequences using MEGA-X, edited and visualized in iTOL. The 316 proteins formed 12 major clades (Fig. 3; color-coded). 11 out of 12 clades (except clade 10) consisted of SAUR homologs from both *A. thaliana* and *B. juncea*; clade 10 contained four SAUR sequences from only *A. thaliana*; two sequences (SAUR 2 / AT4G34780 and SAUR 73 / AT3G03847) did not cluster with any other clades. In the majority of clades, grouping of homologs from a single species was observed as sub-clades. For instance, in clade 3 (green), SAUR 61, 68, 67, 63, 62 and 75 from *A. thaliana* form a sub-clade; whereas the rest of clade 3 consisted of *B. juncea* SAURs forming sub-clades with *A. thaliana* SAURs (e.g., BjSAURA06_2; B08_2; A02-13; B05_12; A03_9; B08_12; A10_2; B02_4 with AtSAUR 39 and AtSAUR 70). Similarly, in clade, 12 (dark blue), SAURs from *B. juncea* formed a large sub-clade (Fig. 3). Within each branch of a sub-clade of *B. juncea* SAURs, orthologs from the A- and B-genome grouped together.

As described previously, we identified a total of 15 conserved amino acid motifs across the *B. juncea* SAUR proteins. When motif distribution was plotted across the phylogenetic tree, homologs that are part of the same sub-clade also showed similarity in motif composition. Motifs 1, 2, and 3 were present across all clades and in most of the SAURs. Along with conserved motifs, clade 2 contains motifs 7, 10, and 12; motifs 8 and 13 were specific to clade 3; motif 15 was observed in clade 5. Motif 4 was found in clades 8, 9, 10, 11, and 12; motif 5 was present in clades 3, 4, and 7; motif 9 was observed in clades 4, 7, 8, 9, 11, and 12 (Fig. 3).

Intra- and Inter-Genomic Synteny Reveals Extent of Segmental and Tandem Duplication

Bi-directional BLAST using *B. juncea* SAUR protein was performed and the results were visualized by CIRCOS plot (Fig. 6A and B; Supplementary Table 10). The analysis identified 53 paralogous gene pairs on A-genome and 73 on B-genome. Detailed analysis of chromosomal coordinates of the query and subject of bi-directional BLAST showed evidence of tandem duplication events on chromosomes B01, B02, B04, B05, and B08, with the maximum number of events on the B05 chromosome (Fig. 6A). In the A-genome, tandem duplication was observed on A02, A03, A09, and A10, with A02 showing the maximum number. No SAUR genes were reported on Chromosome A06 (Fig. 6B).

Analysis of A and B genomes of *B. juncea* revealed that out of the total 106 genes mapped on the A-genome, 31 are related by segmental duplication, 22 are related by tandem duplication and only 2 genes are present in single copies. In the B-genome, out of 130 genes, 43 are related by segmental duplication, 29 are related by tandem duplication and four genes are present as single copy (Supplementary Table 10). Ka/Ks analysis of paralogous pairs with *B. juncea* shows that all the genes are under purifying selection as the Ka/Ks values are < 1 (Supplementary Table 10).

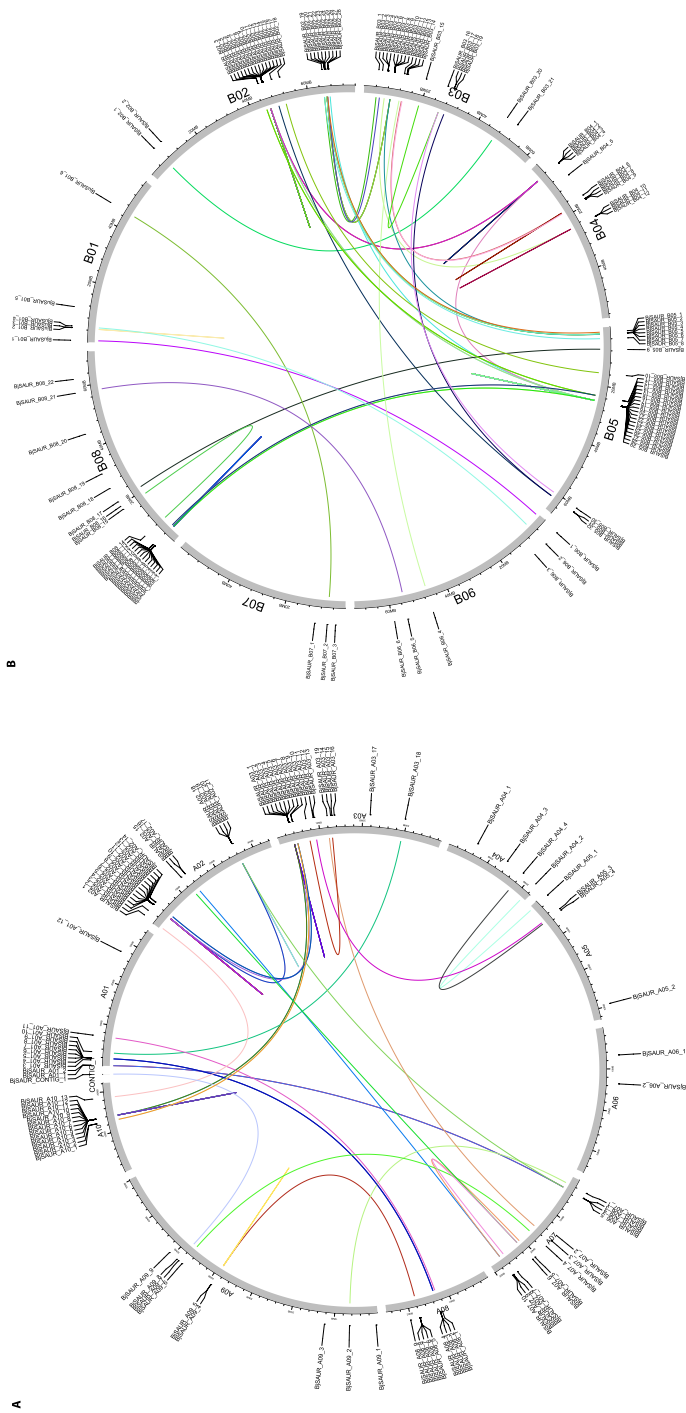


Fig. 6 Synteny of SAUR genes in chromosomes representing A-genome (**A**) and B-genome (**B**) of *B. juncea* visualized using a Circos plot

A clearer and holistic picture of SAUR evolutionary history emerges when phylogenetic reconstruction is correlated with their genomic organisation across the plant family. We therefore performed a comparative analysis of phylogenetic relationship and organisation of SAUR genes that were organised as clusters across selected species of Brassicaceae. The present study identified nine tandemly arranged gene clusters in *A. thaliana* (clusters 1 to 10) based on their chromosomal locations: clusters 1, 2, and 3 are present on chromosome 1; cluster 1 is composed of SAUR 51-52, cluster 2 comprises of SAUR 61-62-63-64-65-66-67; cluster 3 contains SAUR 53-54. Cluster 4 is present on chromosome 2 and comprises SAUR 6-7-12. Cluster 5 (SAUR 26-27-28-29-73) is on chromosome 3; clusters 6 (SAUR 1-2-3-4-5-49-50) and 7 (SAUR 13-14-15-16) are on chromosome 4. Clusters 8 and 9 are on chromosome 5; cluster 8 harbours SAUR 19-20-21-22-23-24; cluster 9 contains SAUR 70-76 (Table 1). Phylogenetic analysis revealed that members of cluster 1 are paralogous to cluster 3; cluster 4 is paralogous to cluster 7, and cluster 5 is paralogous to cluster 8 (Fig. 1B; Supplementary Fig. 2).

To understand whether the organization is of recent origin and limited to *A. thaliana* or is of ancient nature, the organisational pattern of tandemly arranged SAUR genes was explored in selected Brassicaceae species represented by *A. thaliana*, *A. lyrata*, *C. rubella*, *T. halophila*, *C. sativa*, *B. juncea* and *A. arabicum*. A total of 84 SAUR genes were identified in *A. lyrata*, 83 in *C. rubella*, 65 in *T. halophila*, 269 in *C. sativa*, and 28 in *A. arabicum*, besides 79 in *A. thaliana* and 237 in *B. juncea*. Around a 100 kb genomic region flanking the homologous genes was retrieved and global alignment was performed against *A. thaliana* genome (TAIR10) using gVISTA to confirm the presence of SAUR and other flanking genes for each cluster to confirm orthologous nature. *Ab-initio* gene prediction and functional annotation were done using FGENESH and Blast2go (OmicsBox) for synteny analysis (Supplementary Table 11).

In *A. thaliana* cluster 1 comprises tandemly arranged SAUR 54-53 with one intervening gene (Plant protein of unknown function). SAUR 54 and 53 orthologs were found to be conserved among all the selected members of Brassicaceae except in *A. arabicum*, where the ortholog of SAUR 54 was absent. The SAUR 54-53 cluster was detected at a single locus on all species except at two loci in *B. juncea* (B03, B04) and three loci in *C. sativa* (Chr 3, 14, 17). In all the species, SAUR 54 and 53 were disrupted by non-SAUR genes and the number ranged between two in *C. rubella* and *C. sativa* (chr 14) to as many as 9 in *A. lyrata*. In *B. juncea*, the SAUR54-53 clusters on chromosome B03 and B04 are disrupted by 4 and 3 intervening non-SAUR genes, respectively (Fig. 7A; coloured boxes denote SAUR genes). The non-SAUR gene—filament-like plant protein 4 was found to be conserved in *A. lyrata*, *C. rubella*, *T. halophila*, *C. sativa* (chr 3, 14, 17), and *A. arabicum*. Another non-SAUR gene—reverse transcriptase family protein was present only in *A. lyrata*, *C. sativa* (chr 17), and *B. juncea* (B03) (Fig. 7A; black boxes denote non-SAUR genes) (Supplementary Table 11-cluster 1).

Cluster 2 comprises SAUR 61-62-63-64-65-66-67-68 (eight SAUR genes; and three non-SAUR genes present between SAUR 65 and 68; Fig. 8A) in *A. thaliana* on chromosome 1, SAUR 61 to 65 are arranged on the negative strand whereas SAUR 68 to 67 are present on the positive strand. Homologs of the SAUR 61–68

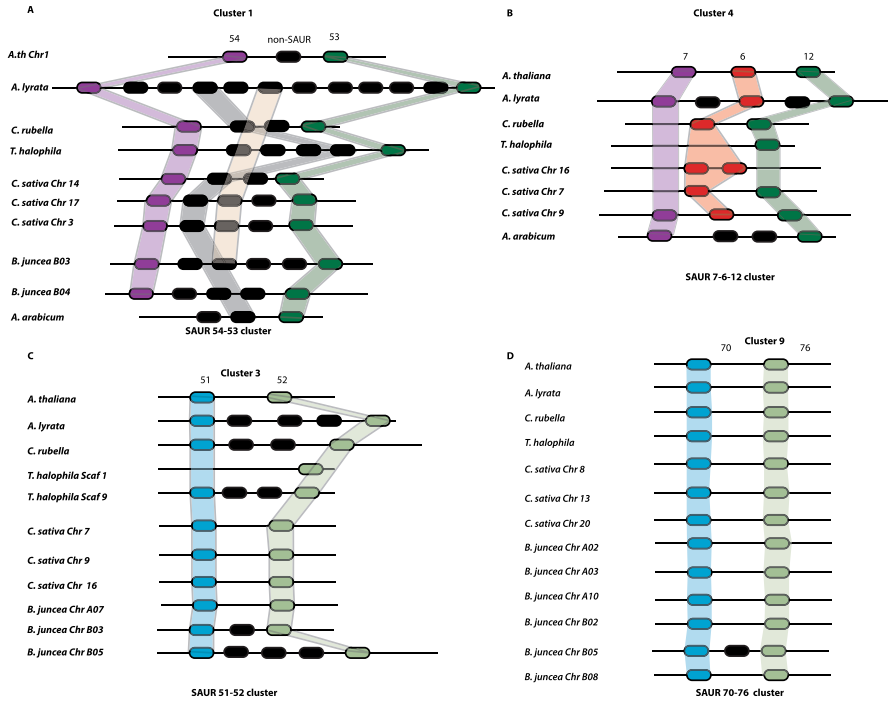


Fig. 7 Schematic representation of micro-synteny of tandemly arranged SAUR gene clusters on pseudochromosomes in selected Brassicaceae members. **A:** Cluster 1 of *AtSAUR* 54–53; **B:** Cluster 4 of *AtSAUR* 7–6–12; **C:** Cluster 3 of *AtSAUR* 51–52; **D:** Cluster 9 of *AtSAUR* 70–76. Similar coloured boxes in each represent orthologous genes, and black boxes represent intervening non-SAUR genes (see supplementary Table 10 for complete annotation) (Color figure online)

cluster were detected on all species that were analysed with a single locus each in *A. lyrata* (seven *SAUR* genes; disrupted at two positions with three non-*SAUR* genes each); *C. rubella* (nine *SAUR* genes, cluster disrupted at two sites, first site with a single, and second site two non-*SAUR* genes), *A. arabicum* (three *SAUR*, disrupted by one non-*SAUR* gene). Clusters of *SAUR* 61–68 homeolog were detected on three chromosomes of *C. sativa* (Chr 3, 14 and 17), and *B. juncea* (A08, A09, B03 and B04). The locus on *C. sativa* chromosome 14 comprises only three *SAUR* genes and eight non-*SAUR* genes; the locus on chr 17 consists of 12 *SAUR* and 14 non-*SAUR* genes; the smallest cluster was on chr 3 with three *SAUR* and one non-*SAUR* gene. The homeologous clusters on *B. juncea* comprised four *SAUR* genes (chr A08), two *SAUR* and three non-*SAUR* (A09), three *SAUR*s (B03) and three *SAUR* and six non-*SAUR* genes (B04) (Fig. 8A). Several non-*SAUR* genes such as *MEI2-like protein 5*, *S-adenosyl-L-methionine-dependent methyltransferases superfamily protein*, *F-box proteins*, *golgin subfamily A member 3* were detected and were found to be variously conserved and duplicated across Brassicaceae (Fig. 8A; Supplementary Table 11-cluster 2).

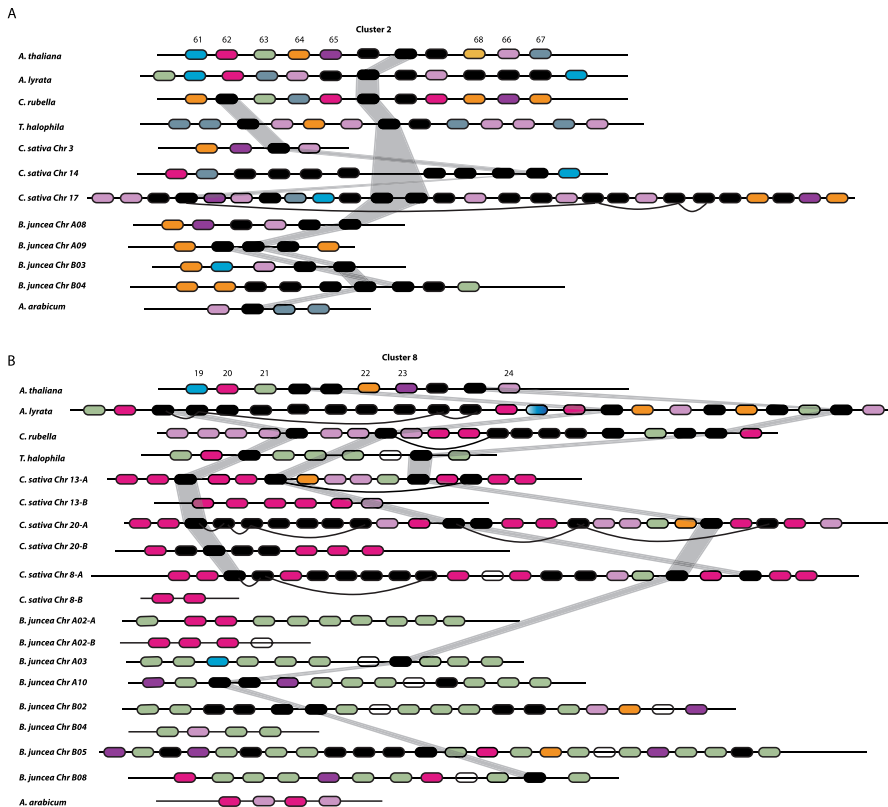


Fig. 8 Schematic representation of micro-synteny analysis of tandemly arranged *SAUR* gene clusters on pseudochromosomes in selected Brassicaceae members. **A** Cluster 2 of *AtSAUR* 61-62-63-64-65-68-66-67; **B** Cluster 8 of *AtSAUR* 19-20-21-22-23-24. Similar coloured boxes in each represent orthologous genes, and black boxes represent intervening non-*SAUR* genes (see supplementary Table 10 for complete annotation) (Color figure online)

In *A. thaliana*, cluster 3 is composed of *SAUR* 51-52 which were found to be tandemly arranged without any intervening gene. In *A. lyrata*, the tandem organisation comprising *SAUR* 51-52 was disrupted by three intervening non-*SAUR* genes, and in *C. rubella*, *SAUR* 51-52 was disrupted by two intervening genes. In *T. halophila*, two loci showed homology to the *SAUR* 51-52 cluster- Scaffold 1 with only *SAUR* 52 ortholog, and Scaffold 9 with *SAUR* 51 and 52 orthologs separated by two non-*SAUR* genes. The *SAUR* 51-52 cluster was present at three loci in *C. sativa* genome, on chromosomes 7, 9, and 16, with all three loci showing tandem organisation of *SAUR* 51-52 without any intervening genes. In *B. juncea*, the *SAUR* 51-52 cluster was also present in multiple copies and on different chromosomes- on A07 as an undisrupted cluster, on B03 with one intervening gene, and on B05 with three intervening genes. No copy of *SAUR* 51-52 was found in *A. arabicum* (Fig. 7C; Supplementary Table 11-cluster 3).

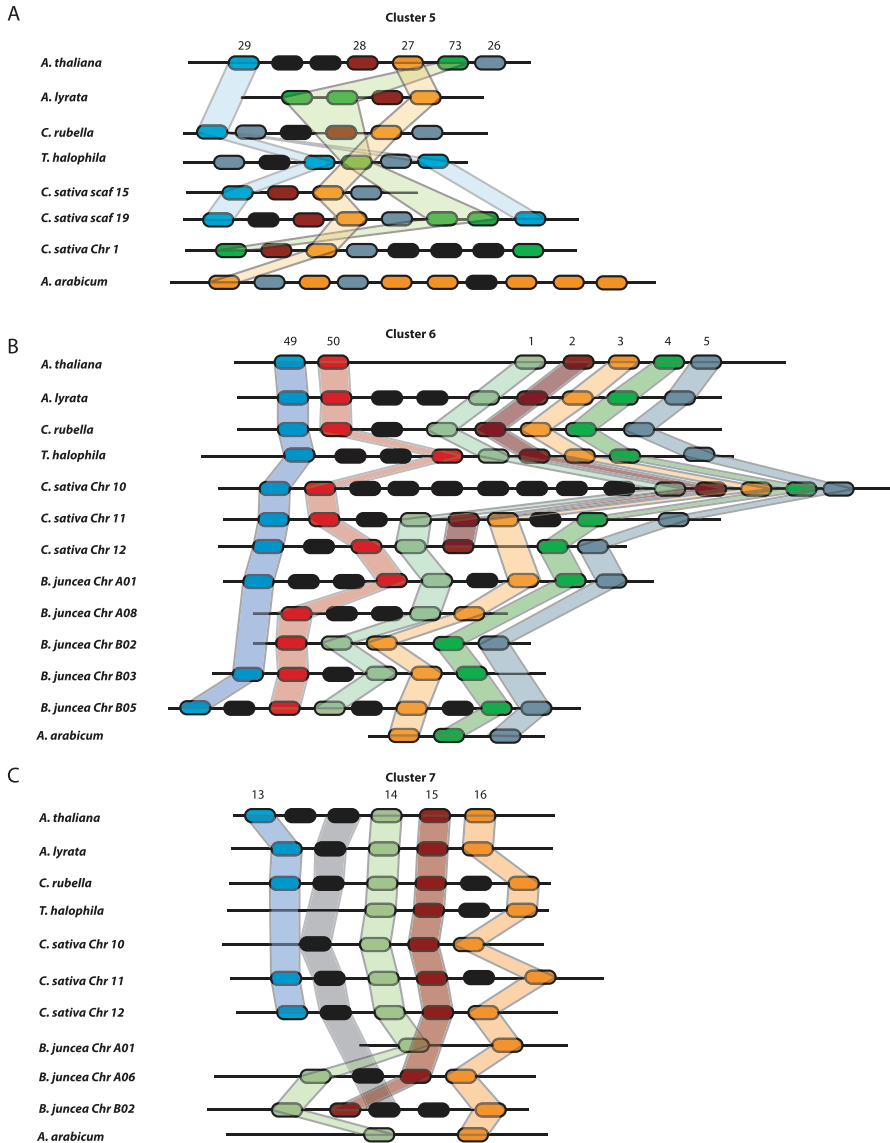


Fig. 9 Schematic representation of micro-synteny analysis of tandemly arranged *SAUR* gene clusters on pseudochromosomes in selected Brassicaceae members. **A** Cluster 5 of *AtSAUR* 29-28-27-73-26; **B** Cluster 6 of *AtSAUR* 49-50-1-2-3-4-5; **C** Cluster 7 of *AtSAUR* 13-14-15-16. Similar coloured boxes in each represent orthologous genes, and black boxes represent intervening non-*SAUR* genes (see supplementary Table 10 for complete annotation) (Color figure online)

In *A. thaliana*, cluster 4 comprises *SAUR* 7-6-12 with no intervening genes. In all the species examined, the cluster was disrupted or partially retained, except on chromosome 9 of *C. sativa*, where a complete and undisrupted cluster was detected.

The entire cluster was found to be deleted from *B. juncea* (Fig. 7B; Supplementary Table 11-cluster 4).

The cluster 5 in *A. thaliana* consists of *SAUR* 29-28-27-73-26, with two intervening genes between *SAUR* 29 and 28. The organization of this cluster was not conserved in its entirety among the Brassicaceae members, as both duplication and deletion were observed. These are exemplified by duplication of *SAUR*73 in *A. lyrata*, *SAUR*29 in *C. rubella* and *T. halophila*, *SAUR* 29 (four copies), *SAUR* 26 (three copies), and two copies of *SAUR* genes which are not part of this cluster) in *A. arabicum*; deletion of *SAUR* 29 and *SAUR* 26 in *A. lyrata*, *SAUR* 73 in *C. rubella*, and *SAUR* 27 in *T. halophila*. In the case of *C. sativa*, three homeologous clusters were identified, and the entire cluster was deleted in *B. juncea* (Fig. 9A). The non-*SAUR* genes were unique to each cluster and loci of different species (Supplementary Table 11-cluster 5).

Cluster 6 comprises *SAUR* 49-50-1-2-3-4-5 without any intervening genes between *SAUR*50 and 1 in the case of *A. thaliana*. Homologs of most of the *SAUR*s were identified, and the organization was found to be conserved among the majority of Brassicaceae members. For instance, two non-*SAUR* genes were present between *SAUR*50 and 1 in *A. lyrata* and one in *C. rubella*; two non-*SAUR*s between *SAUR* 49 and 50 in *T. halophila*. In *A. arabicum*, only *SAUR* 3, 4, and 5 were present. In the two polyploid species- *C. sativa* and *B. juncea*, homeologous clusters were identified. In *C. sativa*, the clusters were localised on chromosomes 10, 11, and 12 with variable numbers of intervening non-*SAUR* genes between different *SAUR* genes (Fig. 9B). In *B. juncea*, five homeologous clusters were observed on chromosomes A01, A08, B02, B03, and B05. Similar to *C. sativa*, duplications, deletions, and variable numbers of non-*SAUR* genes were detected between different *SAUR* genes on the five homeologous clusters (Fig. 9B). For example, in A01cluster, one intervening gene was present between *SAUR* 1 and 3, and *SAUR* 2 was deleted; cluster A08 contains only *SAUR* 50, 1, and 3 with two intervening genes between *SAUR* 50 and 1; cluster B02 contains all *SAUR* genes except *SAUR* 49 and 2; cluster B03 harbours all *SAUR*s except *SAUR* 2 and 5 with one intervening gene between *SAUR* 50 and 1; and, cluster B05 contains all except *SAUR* 2 and one intervening gene each was detected between each *SAUR*3 and 4, *SAUR* 1 and 3, and, *SAUR*49 and 50 (Fig. 9B; Supplementary Table 11-cluster 6). No conserved synteny was observed between the non-*SAUR* genes.

Cluster 7 comprises tandemly arranged *SAUR* 13-14-15-16 with one non-*SAUR* gene intervening *SAUR*13 and 14 in *A. thaliana*. The cluster content and organization were conserved in *A. lyrata*, and with an additional intervening gene between *SAUR* 15 and 16 in *C. rubella*. The cluster was truncated, with only *SAUR*14 and 16 in *A. arabicum*. In *T. halophila*, the cluster comprises *SAUR* 14, 15, and 16 only with an intervening non-*SAUR* gene present between *SAUR*15 and 16. Three homeologous clusters were detected in *C. sativa* (Chromosomes 10, 11, 12), and two homeologous (A01, A05) and one homologous (B02) cluster in *B. juncea*. In *C. sativa*, an intervening gene was present between *SAUR*15 and 16 on chr 11, the cluster on chr 12 was conserved, and *SAUR*13 was deleted from chr 10 cluster. In clusters of *B. juncea*, only *SAUR* 14 and 16 were found on chr A01, *SAUR* 13 was deleted on A06 and B02, and two intervening genes were present between *SAUR* 15

and 16 in the cluster on chromosome B02 (Fig. 9C; Supplementary Table 11-cluster 7). The non-SAUR gene cysteine-rich receptor-like protein kinase 26 was found to be syntenic across most of the species analysed.

Cluster 8, in *A. thaliana*, comprises *SAUR 19-20-21-22-23-24* with two intervening genes each between *SAUR 21-22*, and, *SAUR 23-24*. The cluster composition and organisation were highly variable across all the other members of Brassicaceae and conservation in terms of gene copy number, and order was not observed. For instance, six homologous copies of *SAUR 21* were dispersed on *B. juncea* A02_A and B02, and 11 copies were dispersed on *B. juncea* B05. Similarly, five copies of tandemly arranged homologs of *SAUR 20* were detected on *C. sativa* chr 13-B, and seven dispersed copies on Chr 13-A (Fig. 8B). Among the various non-SAUR genes, phosphoglucosamine mutase-related, NAC (No Apical Meristem) domain transcriptional regulator superfamily protein, and putative F-box protein At1g50870 were present across multiple species (Fig. 8B; Supplementary Table 11-Cluster 8).

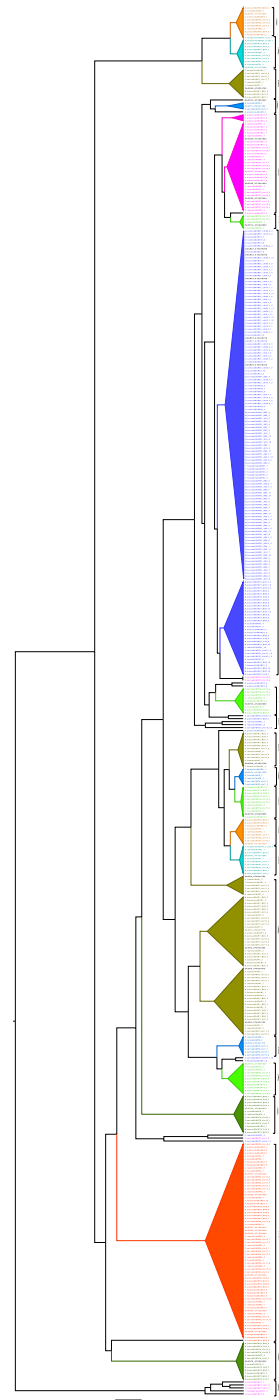
Cluster 9 comprises of *SAUR 70-76* in *A. thaliana*, and this SAUR cluster was found to be the most conserved among all the clusters studied. The cluster was not detected in *A. arabicum*; three homeologous clusters each were identified in *C. sativa* (chromosomes 8, 11, and 20), and of A- genomes (A02, A03, A10) and B- (B02, B08, and B05 with one intervening gene) in *B. juncea* (Fig. 7D; Supplementary Table 11-cluster 9).

Protein sequences of homologs of tandemly arranged *A. thaliana* SAURs from members of Brassicaceae were used to generate a Bayesian-based tree (Fig. 10). Homologs from Brassicaceae corresponding to each of the clusters were found to form single clades. Homologs from clusters 1 and 3, and 6 branched at multiple nodes and formed sister clades, implying paralogy as was observed in *A. thaliana* (Fig. 1B); near similar tree topology was observed between clades 4 and 7, 5 and 8 was observed in *A. thaliana* (Fig. 10).

Discussion

Small auxin up-regulated RNA (SAUR) is the largest among all auxin-inducible gene families, involved in cell growth, elongation, and senescence (Stamm and Kumar 2013; Bemer et al. 2017; van Mourik et al. 2017) and SAUR homologs have been identified and studied from several plants (Markakis et al. 2013; Luan et al. 2023; Wang et al. 2010, 2022, 2020; Bai et al. 2017; Lv et al. 2022; Luo et al. 2022; Hu et al. 2018; Zhang et al. 2023, 2017; Li et al. 2022, 2017; Xie et al. 2015; Chen et al. 2014; Wu et al. 2012). In spite of their widespread occurrence and biological significance, there are no reports on the members of the *SAUR* gene family from *Brassica* species. In addition, the mode of evolution and expansion of this large family especially in the context of polyploidy remains unexplored. We therefore designed a comprehensive *in-silico* analysis schema to identify, annotate, and analyse the organisational pattern of the *SAUR* genes in the allotetraploid species *B. juncea*, its two parents-*B. rapa* and *B. nigra*, and several members of Brassicaceae. *In-silico* analysis revealed 237 *SAUR* homologs in *B. juncea*, 99 in *B. rapa* and 107 in *B.*

Fig. 10 Phylogenetic reconstruction of tandemly arranged *SAUR* gene clusters from *Arabidopsis thaliana* and homologs from Brassicaceae. Nomenclature of clusters is based on *A. thaliana* and represented by different colours. The homologs in each clade confirm the synteny pattern as observed



nigra suggesting an extensive expansion of this family which can be attributed to genome triplication in the ancestral *Brassica* (Cheng et al. 2014; Murat et al. 2015).

The present analysis revealed extensive variation in gene and protein length (4.5-fold), sequence and organisation; the SAUR domain was found to be conserved in all the homologs implying the complexity across members of SAUR proteins. A clear skewness was observed towards fewer introns and these could be classified as being intron-poor (Liu et al. 2021). Such intronless nature of the SAUR genes appears to be widespread as was observed in *O. sativa* (Jain et al. 2006), maize (Chen et al. 2014), Solanaceae (Wu et al. 2012), and *M. domestica* (Wang et al. 2020). The SAUR gene family is likely to have evolved in the last common ancestor of bryophyta (hornworts, liverworts, and mosses), after its divergence from the algal lineage (Bowman et al. 2021). In the earliest examples of land plants, the SAUR gene family numbered between 3 (*Anthoceros augustus*), 14 (*Marchantia polymorpha*) (Bowman et al. 2021) to 15 in *Selaginella moellendorffii* (Zhang et al. 2021). A survey of the SAUR gene family across the green plant lineages reveals a trend of increase in family size, with the majority of the members being intronless (Table 2). The origin of such intronless genes (IGs) is proposed to involve intron splicing in the ancestral copies, reverse transcription, transposition, and duplication events (Lecharny et al. 2003; Chen et al. 2022). Such a preponderance of intronless genes significantly reduces transcriptome diversity by eliminating alternative splicing.

Analysis of the composition of the SAUR homologs in the clusters and their phylogenetic affiliation confirmed that both segmental and tandem duplications and species-specific events are primarily responsible for the expansion of the SAUR family in *Brassica* species, as has been reported for *KCS5-KCS6* (Singh et al. 2018), *MIRNA* genes (Joshi et al. 2018; Bhardwaj et al. 2020). This pattern was consistent with previous studies in rice (Jain et al. 2006); half of the *BhSAUR* genes in wax gourds (*Benincasa hispida*) were presumed to originate via tandem duplication (Luo et al. 2022). In continuation, analysis of selection pressure operating on the paralogs in the *B. juncea* SAUR gene family revealed purifying selection which is essential in maintaining auxin-mediated function also has been demonstrated previously in *Santalum album* (Zhu et al. 2024) and *A. thaliana* (Wang et al. 2019).

Even though the SAUR genes were identified as being early auxin responsive, subsequent studies have uncovered roles of multiple hormones and their cross-talks. For instance, apart from auxin, brassinosteroids and GA positively modulate SAURs, whereas ABA and JA negatively modulate SAURs (Ren and Gray 2015). The present study on promoter analysis allowed us to identify the presence of hormone response elements such as SARE, ABRE, MeJA, and AuxRE in SAUR genes, which are in agreement with several lines of evidence on the role of other hormones such as Brassinosteroids (Yin et al. 2005; Sun et al. 2010; Walcher and Nemhauser 2012), ABA (Kodaira et al. 2011) and TCP regulating the SAUR 63 sub-clade in a GA-dependent manner (Gastaldi et al. 2020).

The precise functions of many of the SAUR genes still remain largely unidentified. Genes within the same clade exhibit high sequence similarity, which is often associated with redundant functions. For instance genes in cluster 9 (*SAUR70*, *76*) primarily influence plant growth by affecting ethylene receptor signalling with some

members also exhibiting calcium binding capacity (Popescu et al. 2007); similarly genes in clusters 5 (*SAUR26-29*) and 8 (*SAUR19-24*) regulate cell expansion through PP2C (Angela et al. 2014). Additionally, genes in cluster 2 (*SAUR61-68*) have been implicated in the stamens and hypocotyls elongation (Chae et al. 2012). Apart from development, SAUR homologs have been identified to play roles in mediating abiotic stress responses. For example, *AtSAUR32* likely mediates drought stress response via the signalling ABA-pathway (He et al. 2021); *OsSAUR32* (from *O. sativa*) in arsenic tolerance in rice (Li et al. 2024); *AtSAUR41* in salt tolerance (Qiu et al. 2020); *AhSAUR3* (from *Arachis hypogaea* L.) and *PtSAUR8* (from *Populus trichocarpa*) in drought tolerance (Liu et al. 2022). Our analysis of SAUR genes in *B. juncea* reveals the presence of SAUR homologs with conserved domains and motifs, suggesting their probable redundant role in plant development and stress response.

Among the 15 motifs that were observed among the *B. juncea* SAUR family, three motifs that correspond to the SAUR domain were present in almost all the SAUR genes suggesting their criticality to function; similar motifs have also been observed in SAUR proteins of *D. longan* (Chen et al. 2023a, b, c), and *S. album* (Zhu et al. 2024).

Prediction of *B. juncea* SAUR proteins localizations are in agreement with what has been reported in *Z. mays* and *P. trichocarpa* (Hu et al. 2018; Chen et al. 2014), with previous reports of localizations in cytoplasm (Kant et al. 2009), nucleus (Park et al. 2007), and plasma membrane (Spartz et al. 2012).

Protein–protein interactome map revealed that SAUR proteins interact with both SAUR and non-SAUR proteins to perform their downstream functions. For instance, a previous study showed auxin-mediated cell elongation occurs through SAUR-PP2C interaction, as also detected in this analysis (Ren et al. 2018). The interaction of SAURs with ARF, BZR, and PIF transcription factors is known to be involved in SAUR-mediated growth response, as shown earlier in *A. thaliana* (Oh et al. 2014; van Mourik et al. 2017). Interestingly, GO analysis revealed that SAUR proteins also have ion and metal binding capacity, which also aligns with the role of calcium ions in auxin signalling through SAUR genes (Vanneste et al. 2013; Yang and Poovaiah 2000).

Mapping of the SAUR genes on the *A. thaliana* genome revealed their genic distribution pattern as tandemly arrayed genes (TAGs). To understand the conservation and divergence of such tandemly arrayed organisation pattern, we analysed a combination of chromosomal localisation, organisation pattern, micro-synteny, and phylogenetic relationship among homologs of the clustered SAUR genes from species across Brassicaceae – viz. *A. lyrata* (Ancestral Crucifer Karyotype-ACK), *C. rubella* (ACK), *C. sativa* (derivative of ACK-dACK or ancestral diploid karyotype-ADK), *T. halophila* (translocated Proto-Calpinieae Karyotype-tPCK), *A. arabicum*, besides *A. thaliana* (reduced/modified ACK) and *B. juncea* (tPCK) (Zhang et al. 2020; Lysak et al. 2005, 2007; Cheng et al. 2017; Mandakova and Lysak 2008). Detailed investigation of the clusters in *A. thaliana* also revealed that several clusters also harbour non-SAUR genes. A comparison of the homologs across members of Brassicaceae, and *A. arabicum* (sister family to Brassicaceae, within Brassicaceae) revealed a similar clustered organisation pattern of the corresponding homologs, which is reflected as

conserved microsynteny. Such tandem array of *SAUR* genes has been reported in *A. thaliana* (Markakis et al. 2013), apple (Wang et al. 2020), cotton (Li et al. 2017), *G. max* (Li et al. 2022), *A. hypogaea* (Liu et al. 2022), and *Eriobotrya japonica* (Gan et al. 2020). Conserved tandem arrayed organisation of *SAUR* genes indicates a common mechanism and ancestral event not only in Brassicaceae but across plant lineages responsible for the initial expansion of the gene family.

Initial analysis of the *A. thaliana* genome had predicted nearly 17% of all genes (4140) being arranged as 1528 tandem arrays (The Arabidopsis Genome Initiative; 2000). Similar estimates of TAGs range between 29% in *O. sativa* (International Rice Genome Sequencing Project. 2005), 15.6% in *P. trichocarpa* (Tuskan et al. 2006), to 35% in *Z. mays* (Messing et al. 2004). An earlier study reported 2,137 TAGs in *B. rapa*, of which 414 were found to be shared with *A. thaliana*, *A. lyrata*, and *T. parvula* (Fang et al. 2012).

Tandemly Arranged Genes (TAGs) are known to originate through local tandem duplications (Stortenbeker and Bemer 2019). Overlaying genomic location information of TAGs with phylogenetic clustering confirmed their paralogous nature. Interestingly, several of the clusters (e.g., clusters 23–20–19–24–2–22 on chromosome 5, cluster 26–28–27–29 on chromosome 3) were found to be present as paralogous segments which arose as a result of segmental duplication. Co-localization of several clusters were found to be conserved between *A. thaliana* and *B. juncea* such as clusters 6 and 7 (Chr. 4; A01, B02), and, 8 and 9 (Chr. 5; A02, A03, A10, B02, B05, B08) indicating large segmental duplications in common ancestral genomes. *SAUR* family thus expanded through a combination of local and segmental duplications, as has been shown earlier for *MIR395* (Rathore et al. 2016; Bhardwaj et al. 2020), *MIR159* (Anand et al. 2019), *MYB42-MYB85* (Yadav et al. 2023). Whether this expansion occurred independently or was an ancestral event will require deeper and global investigations across the plant kingdom, as has been demonstrated for *MIR395* and *MIR399* gene family (Bhardwaj et al. 2020).

Brassicaceae is known to have experienced several whole genome duplications in the common ancestor, as well as in specific lineages such as *Brassica* and *Camelina*. The origin of *B. juncea* is marked not only by allopolyploidization between *B. rapa* (AA) and *B. nigra* (BB) genomes in recent past within as few as ca. 10,000 years ago (Prakash et al. 2009) but also whole genome triplication in the common ancestor of *Brassica* itself, that includes *B. rapa*, *B. nigra* and *B. oleracea* (Cheng et al. 2014). The *B. juncea* genome thus comprises of two-sets of three homeologous copies – Least fractionated (LF), moderate fractionated (MF1), and most fractionated (MF2), each set of LF, MF1, and MF2 from A- and B-genomes (Lysak et al. 2005, 2007; Kang et al. 2021). *B. juncea* thus can be considered a neo-polyploid, that originated because of natural allopolyploidization of two meso-ployploids (He and Bancroft 2018). Homeologous segments harbouring the *SAUR* gene cluster were also identified in *C. sativa*, which has been characterized as a neo-polyploid harbouring three distinct sub-genomes- LF, MF1, and MF2 (Kagale et al. 2014). However, the three distinct sub-genomes may be remnants of the three diploid ancestors – *C. neglecta* (sub-genome 1), and *C. hispida* var. *hispida* and *C. hispida* var. *grandiflora* (sub-genome 3) that are proposed to be ancestors of *C. sativa* (Martin et al. 2022). This complex origin of *B. juncea* was clearly visible in the *SAUR* gene clusters,

as has also been observed in several other gene families, such as Transport inhibitor response 1/auxin signalling *f-box proteins* gene family (TIR1/AFBs) (Cai et al. 2019), *GRAS* gene family (Li et al. 2019), and *CDPK* gene family (Li et al. 2023). However, such expansion events driven by whole genome duplication (WGD) / whole genome triplication (WGT) events are accompanied by extensive losses and local expansions in homeologous segments as was observed in the present study.

Apart from whole-genome-based polyploidization, local events such as single gene- and segmental duplications drive rapid expansion of gene families. Such events can be deciphered by a co-analysis of the genomic / chromosomal location of loci under consideration vis-à-vis their positioning in the phylogenetic clades. Tandem duplications mediated via unequal crossovers play a key role in creating new gene copies within genomic clusters leading to a local increase in copy number and diversity. In contrast, segmental duplications can spread gene copies across different regions of the genome, where they undergo fewer recombination events with the original gene copies (Cannon et al. 2004). Ribosomal genes that are arranged in tandem arrays of thousands of copies are known to maintain a high degree of sequence similarity and homogeneity because of unequal crossing over and concerted evolution (Eickbush and Eickbush 2007; Hemleben et al. 2021; Hillis and Dixon 1991). Whether *SAUR* genes, which albeit are much fewer in number than rDNA genes, residing within or outside the same cluster are also subjected to similar evolutionary processes is yet to be investigated.

An interesting feature of *SAUR* clusters is the presence of non-*SAUR* genes, the majority of which were found to be non-conserved among the orthologous segments across Brassicaceae; only a few, such as *filament-like plant protein 4* (cluster 1), *cysteine-rich receptor-like protein kinase 26* (cluster 7) and *probable methyltransferase PMT24* (cluster 2) were conserved, albeit only in a few species. Such observations are indicative of extensive reorganisation, including loss, retention, and acquisition of non-*SAUR* genes within otherwise conserved *SAUR* gene blocks.

Members of Brassicaceae are valued for several reasons- many species serve as model plants such as *A. thaliana* (model dicot; Woodward and Bartel 2018), *A. lyrata* (reference species for ancestral crucifer karyotype; Murat et al. 2015), *C. rubella* (model for evolution of self-incompatibility; Slotte et al. 2013), *T. halophila* (halophyte model; Wang et al. 2004); whereas several are agronomically useful including *C. sativa* (biofuel; Kagale et al. 2014), and *Brassica* species (oilseeds, vegetables, and condiments; Warwick and Hall 2009). Such extensive morphological and adaptive diversity is attributed to several family- and lineage-specific whole genome duplication events, large segmental, and local duplications (Murat et al. 2015). Duplications lead to the generation of multicopy genes, which in the course of evolution undergo neo-functionalization, sub-functionalisation, or pseudogenization via changes in the coding or regulatory regions (Cheng et al. 2014). Our analysis of the orthologous *SAUR* gene clusters revealed that these were largely conserved among the selected Brassicaceae members; albeit with deletions and duplications. A high degree of conservation was observed among closely related species such as *A. thaliana*—*A. lyrata*—*C. rubella*. In contrast, meso- and neo-polyploid species show comparatively higher gene losses, local duplications, and rearrangements with lesser gene order conservation. For instance, clusters 3 and 9 were not detected

in *A. arabicum*, and clusters 4 and 5 were not detected in *B. juncea*. During the course of evolution, paralogous copies are known to have fully or partially redundant function/s that is necessitated in case of gene loss, as has been observed in the case of *B. juncea* where clusters 4 and 5 were deleted, but their homologous clusters, clusters 7 and 8, respectively, have been retained. In contrast, the organization of clusters 1 (*SAUR* 53–54), 3 (51–52), and 9 (70–76) was found to be highly conserved among the selected Brassicaceae members. Comparative analysis of members of Brassicaceae can thus be useful to understand the association between polyploidy and the evolution of morphological, developmental, regulatory, and adaptive diversity. In addition, comparative analysis can also be deployed to unravel and understand novel regulatory networks involved in development and adaptation (Joshi et al. 2018).

In conclusion, the present study identified the complete compendium of *SAUR* genes in several members of Brassicaceae, most importantly *B. juncea*. The finding lays the framework to not only understand evolutionary forces and trajectories leading to the expansion of gene family, but perform comparative functional genomics to investigate functional and regulatory diversification. The close tandemly arrayed gene cluster organisation also raises significant questions about the inter-relationship between organisational complexity, such as with bidirectional promoter and polycistronic configuration, transcriptional regulation, and functional redundancy and divergence. Future studies on *SAUR* genes in *Brassica* species should be designed to address these biologically relevant questions and to dissect their precise roles in mediating development and stress responses in *B. juncea*.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10528-025-11167-3>.

Acknowledgements This research was supported by DBT, Govt. of India grant BT/PR31101/AGIII/103/1100/2018 to SD. The award of JRF/SRF from CSIR to RS and from UGC to EP is gratefully acknowledged. SD would like to acknowledge Institute of Eminence (Ref. No./IoE/2023-24/12/FRP), and University of Delhi for Infrastructure and financial support.

Author Contributions RS, EP and SD designed the study. RS performed all the analyses (homolog identification, all in-silico analysis) with EP (phylogeny and synteny analysis). RS and SD analyzed all the data and wrote the MS. All the authors reviewed and approved the manuscript.

Funding This research was supported by the DBT, Govt. of India grant BT/PR31101/AGIII/103/1100/2018 to SD. This study was supported by grants from the Institute of Eminence (IoE), Delhi University to SD (Grant Number Ref. No./IoE/2023-24/12/FRP). RS is grateful to CSIR, and EP is grateful to UGC for the award of JRF/SRF.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethical Approval Not applicable.

References

- Anand S, Lal M, Das S (2019) Comparative genomics reveals origin of MIR159A–MIR159B paralogy, and complexities of PTGS interaction between miR159 and target GA-MYBs in Brassicaceae. *Mol Genet Genom* 294:693–714
- Bai MY, Shang JX et al (2012) Brassinosteroid, gibberellin and phytochrome impinge on a common transcription module in *Arabidopsis*. *Nat Cell Biol* 14(8):810–817
- Bai Q, Hou D et al (2017) Genome-wide analysis and expression characteristics of small auxin-up RNA (SAUR) genes in moso bamboo (*Phyllostachys edulis*). *Genome* 60(4):325–336
- Bailey TL, Johnson J, Grant CE, Noble WS (2015) The MEME Suite. *Nucl Acids Res* 43(W1):W39–W49
- Bhardwaj E, Lal M, Anand S, Das S (2020) Independent recurrent evolution of MICRORNA genes converging onto similar non-canonical organisation across green plant lineages is driven by local and segmental duplication events in species, family and lineages. *Plant Sci* 301:110661
- Bowman JL, Flores Sandoval E, Kato H (2021) On the evolutionary origins of land plant auxin biology. *Cold Spring Harb Perspect Biol* 13(6):a040048
- Bray N, Dubchak I, Pachter L (2003) AVID: a global alignment program. *Genome Res* 13(1):97–102
- Bruinsma J, Hasegawa K (1990) A new theory of phototropism—its regulation by a light-induced gradient of auxin-inhibiting substances. *Physiol Planta* 79(4):700–704
- Cannon SB, Mitra A, Baumgarten A et al (2004) The roles of segmental and tandem gene duplication in the evolution of large gene families in *Arabidopsis thaliana*. *BMC Plant Biol* 4:1–21
- Chae K, Isaacs CG, Reeves PH et al (2012) *Arabidopsis* SMALL AUXIN UP RNA63 promotes hypocotyl and stamen filament elongation. *The Plant J* 71(4):684–697
- Chahar N, Dangwal M, Lal M et al (2021) OVATE FAMILY PROTEINS (OFP) gene family across Brassicaceae: Comparative genomic analysis uncovers evolutionary relationships, extensive sequence and structural variation with a potential for functional diversification. *Plant Gene* 28:100343
- Challa KR, Aggarwal P, Nath U (2016) Activation of YUCCA5 by the transcription factor TCP4 integrates developmental and environmental signals to promote hypocotyl elongation in *Arabidopsis*. *Plant Cell* 28(9):2117–2130
- Chen Y, Hao X, Cao J (2014) Small auxin upregulated RNA (SAUR) gene family in maize: identification, evolution, and its phylogenetic comparison with *Arabidopsis*, rice, and sorghum. *J Integr Plant Biol* 56(2):133–150
- Chen F, Wang RJ, Wu CJ et al (2022) SAUR8, a small auxin-up RNA gene in poplar, confers drought tolerance to transgenic *Arabidopsis* plants. *Gene* 837:146692
- Chen Y, Ma T, Zhang T, Ma L (2023a) Trends in the evolution of intronless genes in Poaceae. *Front Plant Sci* 14:1065631
- Chen C, Wu Y, Li J et al (2023b) TBtools-II: a “one for all, all for one” bioinformatics platform for biological big-data mining. *Mole Plant* 16(11):1733–1742
- Chen Y, Ma X, Xue X et al (2023c) Genome-wide analysis of the SAUR gene family and function exploration of DISAUR32 during early longan somatic embryogenesis. *Plant Physiol Biochem* 195:362–374
- Cheng F, Wu J, Wang X (2014) Genome triplication drove the diversification of *Brassica* plants. *Hortic Res* 1:14024
- Cheng F, Liang J, Cai C et al (2017) Genome sequencing supports a multi-vertex model for Brassicaceae species. *Curr Opin Plant Biol* 36:79–87
- Cline M, Wesse T, Iwamura H (1997) Cytokinin/auxin control of apical dominance in *Ipomoea nil*. *Plant Cell Physiol* 38(6):659–667
- Danisman S, Van der Wal F, Dhondt S et al (2012) *Arabidopsis* class I and class II TCP transcription factors regulate jasmonic acid metabolism and leaf development antagonistically. *Plant Physiol* 159(4):1511–1523
- Dong J, Sun N, Yang J et al (2019) The transcription factors TCP4 and PIF3 antagonistically regulate organ-specific light induction of SAUR genes to modulate cotyledon opening during de-etiolation in *Arabidopsis*. *Plant Cell* 31(5):1155–1170
- Drummond AJ, Nicholls GK, Rodrigo AG, Solomon W (2002) Estimating mutation parameters, population history and genealogy simultaneously from temporally spaced sequence data. *Genet* 161:1307–1320
- Drummond AJ, Suchard MA, Xie D, Rambaut A (2012) Bayesian phylogenetics with BEAUti and the BEAST 1.7. *Mol Biol Evol* 29(8):1969–1973

- Edgar RC (2004) MUSCLE: a multiple sequence alignment method with reduced time and space complexity. *BMC Bioinform* 5:113
- Eickbush TH, Eickbush DG (2007) Finely orchestrated movements: evolution of the ribosomal RNA genes. *Genet* 175(2):477–485
- Fang L, Cheng F, Wu J, Wang X (2012) The impact of genome triplication on tandem gene evolution in *Brassica rapa*. *Front Plant Sci* 3:261
- Franklin KA, Lee SH, Patel D et al (2011) Phytochrome-interacting factor 4 (PIF4) regulates auxin biosynthesis at high temperature. *Proc Natl Acad Sci USA* 108(50):20231–20235
- Franzke A, Lysak MA, Al-Shehbaz IA et al (2011) Cabbage family affairs: the evolutionary history of Brassicaceae. *Trends Plant Sci* 16(2):108–116
- Gan X, Jing Y, Shahid MQ et al (2020) Identification, phylogenetic analysis, and expression patterns of the SAUR gene family in loquat (*Eriobotrya japonica*). *Turk J Agri Forestry* 44(1):15–23
- Gastaldi V, Lucero LE, Ferrero LV et al (2020) Class-I TCP transcription factors activate the SAUR63 gene subfamily in gibberellin-dependent stamen filament elongation. *Plant Physiol* 182(4):2096–2110
- Gasteiger E, Hoogland C, Gattiker A et al (2005) Protein identification and analysis tools on the ExPASy server. In: Walker JM (ed) The proteomics protocols handbook: springer protocols handbooks. Humana Press, pp 571–607
- Hagen G, Guilfoyle T (2002) Auxin-responsive gene expression: genes, promoters and regulatory factors. *Plant Mol Biol* 49:373–385
- He Z, Bancroft I (2018) Organization of the genome sequence of the polyploid crop species *Brassica juncea*. *Nat Genet* 50:1496–1497
- He Y, Liu Y, Li M et al (2021) The *Arabidopsis* SMALL AUXIN UP RNA32 protein regulates ABA-mediated responses to drought stress. *Front Plant Sci* 12:625493
- Hemleben V, Grierson D, Borisjuk N et al (2021) Personal perspectives on plant ribosomal rna genes research: from precursor-rRNA to molecular evolution. *Front Plant Sci* 12:797348
- Hillis DM, Dixon MT (1991) Ribosomal DNA: molecular evolution and phylogenetic inference. *Q Rev Biol* 66(4):411–453
- Horton P, Park KJ, Obayashi T et al (2007) WoLF pSORT: protein localization predictor. *Nucl Acids Res* 35(2):W585–W587
- Hou QD, Hong Yi, Zhuang WEN (2023) Molecular characterization of the SAUR gene family in sweet cherry and functional analysis of PavSAUR55 in the process of abscission. *J Int Agri* 22(6):1720–1739
- Hu B, Jin J, Guo AY et al (2015) GSDS 20: an upgraded gene feature visualization server. *Bioinform* 31(8):1296–1297
- Hu W, Yan H, Luo S, Pan F, Wang Y, Xiang Y (2018) Genome-wide analysis of poplar SAUR gene family and expression profiles under cold, polyethylene glycol and indole-3-acetic acid treatments. *Plant Physiol Biochem* 128:50–65
- Istace B, Belser C, Falentin C et al (2021) Sequencing and chromosome-scale assembly of plant genomes, *Brassica rapa* as a use case. *Biology* 10(8):732
- Jain M, Tyagi AK, Khurana JP (2006) Genome-wide analysis, evolutionary expansion, and expression of early auxin-responsive SAUR gene family in rice (*Oryza sativa*). *Genomics* 88(3):360–371
- Jia C, Shi Y, Wang H et al (2024) Genome-wide identification and expression analysis of SMALL AUXIN UP RNA (SAUR) genes in rice (*Oryza sativa*). *Plant Signal Behav* 19(1):2391658
- Johnson MA, Pérez-Amador M A, Lidder P, Green PJ (2000) Mutants of *Arabidopsis* defective in a sequence-specific mRNA degradation pathway. *Proc Natl Acad Sci USA* 97(25):13991–13996
- Joshi G, Chauhan C, Das S (2018) Microsynteny analysis to understand evolution and impact of polyploidization on MIR319 family within Brassicaceae. *Dev Genes Evol* 228(6):227–242
- Kagale S, Robinson SJ, Nixon J, Xiao R, Huebert T (2014) Polyploid evolution of the Brassicaceae during the Cenozoic era. *Plant Cell* 26(7):2777–2791
- Kang L, Qian L, Zheng M, Chen L, Chen H (2021) Genomic insights into the origin, domestication and diversification of *Brassica juncea*. *Nat Genet* 53(9):1392–1402
- Kant S, Bi YM, Zhu T, Rothstein SJ (2009) SAUR39, a small auxin-up RNA gene, acts as a negative regulator of auxin synthesis and transport in rice. *Plant Physiol* 151(2):691–701
- Kodaira KS, Qin F, Tran LSP, Maruyama K, Kidokoro S et al (2011) *Arabidopsis* Cys2/His2 zinc-finger proteins AZF1 and AZF2 negatively regulate abscisic acid-repressive and auxin-inducible genes under abiotic stress conditions. *Plant Physiol* 157(2):742–756

- Kumar S, Stecher G, Li M, Knyaz C, Tamura K (2018) MEGA X: molecular evolutionary genetics analysis across computing platforms. *Mol Biol Evol* 35(6):1547–1549
- Lal M, Bhardwaj E, Chahar N, Yadav S, Das S (2022) Comprehensive analysis of 1R-and 2R-MYBs reveals novel genic and protein features, complex organisation, selective expansion and insights into evolutionary tendencies. *Funct Integ Genom* 22(3):371–405
- Lecharny A, Boudet N, Gy I, Aubourg S, Kreis M (2003) Introns-in, introns-out in plant gene families: a genomic approach of the dynamics of gene structure. *J Struct Funct Genom* 3:111–116
- Li X, Liu G, Geng Y, Wu M, Pei W, Zhai H, Zang X, Li X, Zhang J, Yu S, Yu J (2017) A genome-wide analysis of the small auxin-up RNA (SAUR) gene family in cotton. *BMC Genomics* 18(1):815
- Li M, Sun B, Xie F, Gong R, Luo Y, Zhang F, Tang H (2019) Identification of the GRAS gene family in the *Brassica juncea* genome provides insight into its role in stem swelling in stem mustard. *PeerJ* 7:e6682
- Li G, Wang Q, Lu L, Wang S, Chen X, Khan MHU, Yang S (2022) Identification of the soybean small auxin upregulated RNA (SAUR) gene family and specific haplotype for drought tolerance. *Biologia* 77(4):1197–1217
- Li H, Wu H, Huang W, Liu J, Deng J, Li C, Zheng J (2023) The CDPK gene family in mustard (*Brassica juncea* L.): genome-wide identification and expression analysis under cold stress. *Horticulturae* 10(1):20
- Li C, Jiang C, Pan X, Qi Y, Zhao W, Dong H, Liu Q (2024) OsSAUR2, a small auxin-up RNA gene, is crucial for arsenic tolerance and accumulation in rice. *Environ Exp Bot* 226:105894
- Liu H, Lyu HM, Zhu K, Van de Peer Y, Max Cheng ZM (2021) The emergence and evolution of intron-poor and intronless genes in intron-rich plant gene families. *Plant J* 105(4):1072–1082
- Liu Y, Xiao L, Chi J, Li R, Han Y, Cui F, Li G (2022) Genome-wide identification and expression of SAUR gene family in peanut (*Arachis hypogaea* L.) and functional identification of AhSAUR3 in drought tolerance. *BMC Plant Biol* 22(1):178
- Luan J, Xin M, Qin Z (2023) Genome-wide identification and functional analysis of the roles of SAUR gene family members in the promotion of cucumber root expansion. *Int J Mol Sci* 24(6):5940
- Luo C, Yan J, He C, Liu W, Xie D, Jiang B (2022) Genome-wide identification of the SAUR gene family in wax gourd (*Benincasa hispida*) and functional characterization of BhSAUR60 during fruit development. *Int J Mol Sci* 23(22):14021
- Lv W, He X, Guo H, Lan H, Jiao Y, Li L, Lin T (2022) Genome-wide identification of taSAUR gene family members in hexaploid wheat and functional characterization of taSAUR66-5B in improving nitrogen use efficiency. *Int J Mol Sci* 23(14):7574
- Lysak MA, Koch MA, Pecinka A, Schubert I (2005) Chromosome triplication found across the tribe Brassiceae. *Genome Res* 15(4):516–525
- Lysak MA, Cheung K, Kutschke M, Bures P (2007) Ancestral chromosomal blocks are triplicated in Brassiceae species with varying chromosome number and genome size. *Plant Physiol* 145(2):402–410
- Ma X, Dai S, Qin N, Zhu C, Qin J, Li J (2023) Genome-wide identification and expression analysis of the SAUR gene family in foxtail millet (*Setaria italica* L.). *BMC Plant Biol* 23(1):31
- Mandáková T, Pouch M, Brock JR, Al-Shehbaz IA, Lysak MA (2019) Origin and evolution of diploid and allopolyploid Camelina genomes were accompanied by chromosome shattering. *Plant Cell* 31(11):2596–2612
- Mandakova T, Lysak MA (2008) Chromosomal phylogeny and karyotype evolution in x= 7 crucifer species (Brassicaceae). *Plant Cell* 20(10):2559–2570
- Markakis MN, Boron AK, Van Loock B, Saini K, Cirera S, Verbelen JP, Vissenberg K (2013) Characterization of a small auxin-up RNA (SAUR)-like gene involved in *Arabidopsis thaliana* development. *PLoS ONE* 8(11):e82596
- Martin SL, Lujan Toro B, James T, Sauder CA, Laforest M (2022) Insights from the genomes of 4 diploid Camelina spp. G3 (Bethesda). 12(12):jkac182
- McClure BA, Guilfoyle T (1987) Characterization of a class of small auxin-inducible soybean polyadenylated RNAs. *Plant Mol Biol* 9:611–623
- Messing J, Bharti AK, Karlowski WM, Gundlach H, Kim HR, Yu Y, Wei F, Fuks G, Soderlund CA, Mayer KF, Wing RA (2004) Sequence composition and genome organization of maize. *Proc Natl Acad Sci USA* 101(40):14349–14354
- Murat F, Louis A, Maumus F (2015) Understanding Brassicaceae evolution through ancestral genome reconstruction. *Genome Biol* 16:262
- Nagaharu U (1935) Genome analysis in Brassica with special reference to the experimental formation of *B. napus* and peculiar mode of fertilization Japan. *J Bot* 7:389–452

- Nemhauser JL, Hong F, Chory J (2006) Different plant hormones regulate similar processes through largely nonoverlapping transcriptional responses. *Cell* 126(3):467–475
- Oh E, Zhu JY, Bai MY, Arenhart RA, Sun Y, Wang ZY (2014) Cell elongation is regulated through a central circuit of interacting transcription factors in the *Arabidopsis* hypocotyl. *Elife* 3:e03031
- Park JE, Kim YS, Yoon HK, Park CM (2007) Functional characterization of a small auxin-up RNA gene in apical hook development in *Arabidopsis*. *Plant Sci* 172(1):150–157
- Paysan-Lafosse T, Blum M, Chuguransky S, Grego T, Pinto BL, Salazar GA, Bateman A (2023) InterPro in 2022. *Nucleic Acids Res* 51(D1):D418–D427
- Perumal S, Koh CS, Jin L, Buchwaldt M, Higgins EE, Zheng C, Sankoff D, Robinson SJ, Kagale S, Navabi ZK, Tang L, Horner KN, He Z, Bancroft I, Chalhoub B, Sharpe AG, Parkin IAP (2020) A high-contiguity *Brassica nigra* genome localizes active centromeres and defines the ancestral Brassica genome. *Nat Plants* 6(8):929–941
- Popescu SC, Popescu GV, Bachan S, Zhang Z, Seay M, Gerstein M, Dinesh-Kumar SP (2007) Differential binding of calmodulin-related proteins to their targets revealed through high-density *Arabidopsis* protein microarrays. *Proc Natl Acad Sci* 104(11):4730–4735
- Prakash S, Bhat SR, Quiros CF, Kirti PB, Chopra VL (2009) 2 Brassica and its close allies: cytogenetics and evolution. *Plant Breed Rev* 31(23):21–187
- Pybus OG, Rambaut A (2002) GENIE: estimating demographic history from molecular phylogenies. *Bioinformatics* 18:1404–1405
- Qiao X, Li Q, Yin H, Qi K, Li L, Wang R, Zhang S, Paterson AH (2019) Gene duplication and evolution in recurring polyploidization-diploidization cycles in plants. *Genome Biol* 20(1):38
- Qiu T, Qi M, Ding X, Zheng Y, Zhou T, Chen Y, Han N, Zhu M, Bian H, Wang J (2020) The SAUR41 subfamily of SMALL AUXIN UP RNA genes is abscisic acid inducible to modulate cell expansion and salt tolerance in *Arabidopsis thaliana* seedlings. *Ann Bot* 125(5):805–819
- Rambaut A (2000) Estimating the rate of molecular evolution: incorporating non-contemporaneous sequences into maximum likelihood phylogenies. *Bioinformatics* 16:395–399
- Rambaut A. 2018. FigTree. Version 1.4.4. <http://tree.bio.ed.ac.uk/software/figtree/>.
- Rathore P, Geeta R, Das S (2016) Microsynteny and phylogenetic analysis of tandemly organised miRNA families across five members of Brassicaceae reveals complex retention and loss history. *Plant Sci* 247:35–48
- Ren H, Gray WM (2015) SAUR proteins as effectors of hormonal and environmental signals in plant growth. *Mol Plant* 8(8):1153–1164
- Ren H, Park MY, Spartz AK, Wong JH, Gray WM (2018) A subset of plasma membrane-localized PP2C: D phosphatases negatively regulate SAUR-mediated cell expansion in *Arabidopsis*. *PLoS Genet* 14(6):e1007455
- Serrani JC, Ruiz-Rivero O, Fos M, García-Martínez JL (2008) Auxin-induced fruit-set in tomato is mediated in part by gibberellins. *Plant J* 56(6):922–934
- Shin J-H, Mila I, Liu M, Rodrigues MA, Vernoux T, Pirrello J, Bouzayen M (2019) The RIN-regulated Small Auxin-Up RNA SAUR69 is involved in the unripe-to-ripe phase transition of tomato fruit via enhancement of the sensitivity to ethylene. *New Phytol* 222(2):820–836
- Slotte T, Hazzouri K, Ågren J et al (2013) The *Capsella rubella* genome and the genomic consequences of rapid mating system evolution. *Nat Genet* 45:831–835
- Solovyev V, Kosarev P, Seledsov I, Vorobyev D (2006) Automatic annotation of eukaryotic genes, pseudogenes and promoters. *Genome Biol* 7:1–12
- Spartz AK, Lee SH, Wenger JP, Gonzalez N, Itoh H, Inzé D, Gray WM (2012) The SAUR19 subfamily of SMALL AUXIN UP RNA genes promote cell expansion. *Plant J* 70(6):978–990
- Spartz AK, Ren H, Park MY, Grandt KN, Lee SH, Murphy AS, Gray WM (2014) SAUR inhibition of PP2C-D phosphatases activates plasma membrane H⁺-ATPases to promote cell expansion in *Arabidopsis*. *Plant Cell* 26(5):2129–2142
- Stamm P, Kumar PP (2013) Auxin and gibberellin responsive *Arabidopsis* SMALL AUXIN UP RNA36 regulates hypocotyl elongation in the light. *Plant Cell Rep* 32:759–769
- Stortenbeker N, Bemer M (2019) The SAUR gene family: the plant's toolbox for adaptation of growth and development. *J Exp Bot* 70(1):17–27
- Suchard MA, Lemey P, Baele G, Ayres DL, Drummond AJ & Rambaut A (2018) Bayesian phylogenetic and phylodynamic data integration using BEAST 1.10 *Vir Evol* 4: vey016.
- Sun Y, Fan XY, Cao DM, Tang W, He K, Zhu JY, Wang ZY (2010) Integration of brassinosteroid signal transduction with the transcription network for plant growth regulation in *Arabidopsis*. *Devel Cell* 19(5):765–777

- Teale WD, Paponov IA, Palme K (2006) Auxin in action: signalling, transport and the control of plant growth and development. *Nat Rev Mol Cell Biol* 7(11):847–859
- Tian Z, Han J, Che G, Hasi A (2022) Genome-wide characterization and expression analysis of SAUR gene family in Melon (*Cucumis melo* L.). *Planta* 255(6):123
- Tiley GP, Ané C, Burleigh JG (2016) Evaluating and characterizing ancient whole-genome duplications in plants with gene count data. *Genom Biol Evol* 8(4):1023–1037
- Tuskan GA, Difazio S, Jansson S, Bohlmann J, Grigoriev I, Hellsten U, Rokhsar D (2006) The genome of black cottonwood, *Populus trichocarpa* (Torr. and Gray). *Science* 313(5793):1596–1604
- Voorrips R (2002) MapChart: software for the graphical presentation of linkage maps and QTLs. *J Hered* 93(1):77–78
- Walcher CL, Nemhauser JL (2012) Bipartite promoter element required for auxin response. *Plant Physiol* 158(1):273–282
- Wang S, Bai Y, Shen C, Wu Y, Zhang S, Jiang D, Guilfoyle TJ, Chen M, Qi Y (2010) Auxin-related gene families in abiotic stress response in *Sorghum bicolor*. *Funct Integr Genom* 10(4):533–546
- Wang Z, Yang L, Liu Z, Lu M, Wang M, Sun Q, Hua J (2019) Natural variations of growth thermo-responsiveness determined by SAUR 26/27/28 proteins in *Arabidopsis thaliana*. *New Phytol* 224(1):291–305
- Wang P, Lu S, Xie M, Wu M, Ding S, Khaliq A, Ma Z, Mao J, Chen B (2020) Identification and expression analysis of the small auxin-up RNA (SAUR) gene family in apple by inducing of auxin. *Gene* 750:144725
- Wang M, Manzoora MA, Wang X, Feng X, Zhao Y, He J, Cai Y (2022) Comparative genomic analysis of SAUR gene family, cloning and functional characterization of two genes (*PbrSAUR13* and *PbrSAUR52*) in *Pyrus bretschneideri*. *Int J Mol Sci* 23(13):7054
- Wang Y, Jia L, Tian G, Dong Y, Zhang X, Zhou Z, Yao W (2023) shinyCircos-V2.0: leveraging the creation of Circos plot with enhanced usability and advanced features. *Imeta* 2(2):e109
- Warwick SI, Hall JC (2009) Phylogeny of *Brassica* and wild relatives. *Biol Breed Cruci* 19:36
- Woodward AW, Bartel B (2005) Auxin: regulation, action, and interaction. *Ann Bot* 95(5):707–735
- Woodward AW, Bartel B (2018) Biology in bloom: a primer on the *Arabidopsis thaliana* model system. *Genetics* 208(4):1337–1349
- Wu J, Liu S, He Y, Guan X, Zhu X, Cheng L, Wang J, Lu G (2012) Genome-wide analysis of SAUR gene family in Solanaceae species. *Gene* 509(1):38–50
- Xie R, Dong C, Ma Y, Deng L, He S, Yi S, Zheng Y (2015) Comprehensive analysis of SAUR gene family in citrus and its transcriptional correlation with fruitlet drop from abscission zone A. *Funct Integr Genom* 15:729–740
- Yadav S, Chahar N, Lal M, Das S (2023) Phylogenetic and comparative genomics establishes origin of paralogy between homologs of *AtMYB42* and *AtMYB85* in last common ancestor of Brassicaceae via segmental duplication. *Plant Gene* 35:100424
- Yang T, Poovaiah BW (2000) Molecular and biochemical evidence for the involvement of calcium/calmodulin in auxin action. *J Biol Chem* 275(5):3137–3143
- Yang J, Liu D, Wang X, Ji C, Cheng F, Liu B, Zhang M (2016) The genome sequence of allopolyploid *Brassica juncea* and analysis of differential homoeolog gene expression influencing selection. *Nat Genet* 48(10):1225–1232
- Yin Y, Vafeados D, Tao Y, Yoshida S, Asami T, Chory J (2005) A new class of transcription factors mediates brassinosteroid-regulated gene expression in *Arabidopsis*. *Cell* 120(2):249–259
- Zhang N, Huang X, Bao Y, Wang B, Zeng H, Cheng W, Sun Y (2017) Genome-wide identification of SAUR genes in watermelon (*Citrullus lanatus*). *Physiol Mol Biol Plants* 23:619–628
- Zhang Z, Meng F, Sun P, Yuan J, Gong K, Liu C, Wang X (2020) An updated explanation of ancestral karyotype changes and reconstruction of evolutionary trajectories to form *Camelina sativa* chromosomes. *BMC Genom* 21:1–9
- Zhang H, Yu Z, Yao X, Chen J, Chen X, Zhou H, Lou Y, Ming F, Jin Y (2021) Genome-wide identification and characterization of small auxin-up RNA (SAUR) gene family in plants: evolution and expression profiles during normal growth and stress response. *BMC Plant Biol* 21(1):4
- Zhang Y, Ye T, She Z, Huang S, Wang L, Aslam M, Niu X (2023) Small Auxin Up RNA (SAUR) gene family identification and functional genes exploration during the floral organ and fruit developmental stages in pineapple (*Ananas comosus* L.) and its response to salinity and drought stresses. *Int J Biol Macro* 237:124061
- Zhu Q, Zheng H, Hu X, Liu Y, Zheng X, Li L, Tang M (2024) Genome-wide analysis of the SAUR gene family and its expression profiles in response to salt stress in *Santalum album*. *Plants* 13(10):128

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.